



TRAINER SLIDES · WSQ

Effective Project Management for Small Projects

WSQ Course Code: TGS-2020505545 ·

2 days · 16 training hours

TSC: Project Management · ICT-PMT-3026-1.1

Version v19.0 · 19 August 2026

Tertiary Infotech Academy Pte Ltd · UEN: 201200696W



COURSE ADMINISTRATION

Welcome & Housekeeping

How the two days work and what evidence you will produce

Digital Attendance (Mandatory)

AM, PM and Assessment

Complete every required SSG digital attendance scan.

Scan the trainer QR

Use your mobile camera and submit the attendance record.

75% minimum

Attendance eligibility is calculated from the official record.

Assessment required

Complete the assessment and achieve Competent for funding eligibility.

Your Trainer

Name

Dr Alfred Ang

Role

Course developer and principal trainer, Tertiary Infotech Academy

Experience

20+ years across engineering, analytics, AI and digital transformation

Contact

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Ground Rules

Participate

Use the tools, challenge assumptions and explain your decisions.

Respect

One conversation at a time; disagree with evidence, not people.

Be punctual

Activities build on one another; missed work creates downstream gaps.

Keep evidence

Save every completed template for the case study and assessment review.

Learning Outcomes — 1 to 4

LO1

Develop project scope and team to initiate the facilitation of a small project

LO2

Develop a realistic project plan based on the agreed project scope

LO3

Utilise appropriate methods and tools to track and drive a project

LO4

Identify, analyse and manage project risks

Learning Outcomes — 5 to 7

LO5

Collaborate effectively with project stakeholders

LO6

Deploy manpower, budget and other resources for efficient completion

LO7

Track project deliverables and correct misalignments through control and closeout

Skills Framework — Abilities A1 to A4

Code	Ability	Practice
A1	Facilitate execution of small projects that are typically less than six months, with limited budget and interdependencies, and no significant strategic impact	Activities 1–2
A2	Implement realistic project plans based on project objectives and scope	Activities 3–5
A3	Utilise appropriate methods and tools to track and drive progress against plans and timelines	Activities 6 and 10
A4	Identify risks to project success and take appropriate actions to manage them	Activity 7

Every assessed ability has named tool-based evidence across the lifecycle.

Skills Framework — Abilities A5 to A7

Code	Ability	Practice
A5	Collaborate effectively with relevant internal and external stakeholders directly impacting the project	Activity 8
A6	Deploy manpower, financial budgets and relevant resources for efficient and effective completion	Activity 9
A7	Track deliverables against schedules, monitor costs, timescales and resources, and take basic corrective actions when misaligned	Activities 11–12

The twelve activities form one connected project evidence trail.

Skills Framework — Knowledge

Code	Knowledge	Taught in
K1	Elements of a small project	Topic 1
K2	Requirements of a project plan	Topic 2
K3	Application of appropriate project management methodologies and tools	Topic 3
K4	Project risks	Topic 4
K5	Project stakeholder identification	Topic 5

Knowledge is applied in the activity immediately following its topic.

Lesson Plan — Day 1

Initiate with value

Topic 1 · charter, boundaries, team and kickoff

Plan the work

Topic 2 · requirements, WBS, schedule, critical path and budget

Drive quality and flow

Topic 3 · visual management, root-cause and prioritisation tools

Manage uncertainty

Topic 4 · risk register, responses, contingency and decision analysis

Lesson Plan — Day 2

Engage stakeholders

Topic 5 · mapping, RACI, communication and engagement

Deploy resources

Topic 6 · capacity, optimisation, procurement and team performance

Control and close

Topic 7 · status, earned value, change, corrective action and handover

Assessment

Written Case Study and Oral Questioning, open book

Course Outline — Topics 1 to 4

Topic 01 · Initiation of a Project

Small-project fit · value · charter · scope · team · kickoff · A1, K1

Topic 02 · Develop a Project Plan

Requirements · scope baseline · WBS · network · schedule · budget · A2, K2

Topic 03 · Apply Methods and Tools to Drive a Project

Quality planning · flow · root cause · Pareto · improvement · A3, K3

Topic 04 · Manage Project Risk

Identify · assess · respond · contingency · monitor · A4, K4

Course Outline — Topics 5 to 7

Topic 05 · Collaborate with Stakeholders

Identify · analyse · communicate · engage · allocate accountability · A5, K5

Topic 06 · Deploy Resources

People · budget · equipment · procurement · team performance · A6

Topic 07 · Track Project Deliverables and Correct Misalignments

Performance · earned value · change · corrective action · handover · closure · A7

The Twelve Tool Packs

Activity 1

Selection & Cost-Benefit — Screen and select the project

Activity 2

Business Case & Charter — Authorise and kick off

Activity 3

Requirements Workshop & RTM — Discover and trace needs

Activity 4

WBS & Dictionary — Build the scope baseline

Activity 5

Network, Gantt & Budget — Integrate schedule and cost

Activity 6

Delivery & Quality Plan — Tailor execution and quality

Activity 7

Risk Decision Tools — Fund risk responses

Activity 8

Stakeholder, RACI & SEAM — Align decisions and engagement

Activity 9

Capacity & Procurement — Deploy resources

Activity 10

Root-Cause Toolchain — Diagnose quality

Activity 11

Status, EVM, Issue & Change — Control performance

Activity 12

Acceptance, Handover & Benefits — Close and transfer

One Small Project Runs Through All Twelve Activities

Scope

A three-outlet click-and-collect pilot for HarbourHub Retail.

Time & money

Five months, a S\$180,000 cap and a seven-person core team.

Interfaces

Two vendors plus outlet operations, finance, IT and customers.

Success

Launch on date; 120 orders/week; 98% accuracy; 4.2/5 satisfaction; within cap.

Briefing for Assessment

Individual work

No discussion once assessment begins.

Open book

Slides, Learner Guide and approved materials only.

Protect the paper

No photographs, recording or unauthorised materials.

Answer with evidence

State the method, decision, rationale and expected result.

Final Assessment

Written Case Study

Written Assessment (Case Study) — apply project management judgement to a small-project scenario.

Oral Questioning

Oral Questioning — explain and justify decisions against the course methods and evidence.

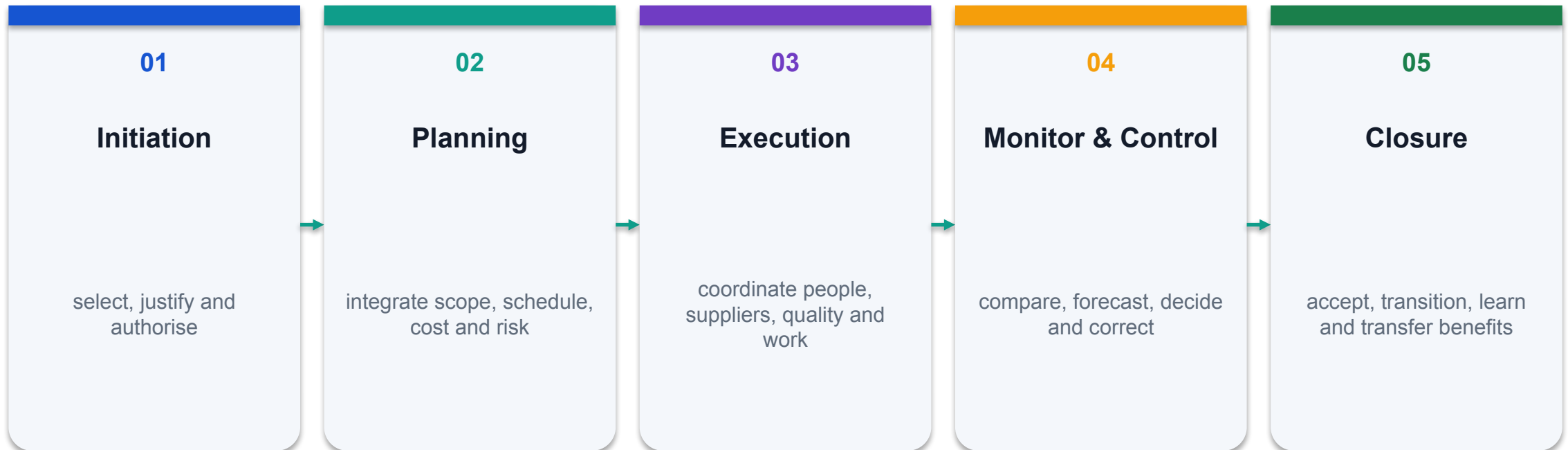
Conditions

Open book: course slides, Learner Guide and approved course materials only.

Competency

A minimum of 75% attendance and a Competent result are required for funding eligibility.

The Complete Project Lifecycle



The phases overlap in practice; control and engagement operate from authorisation to closure.

Lifecycle Phases and WSQ Topics

Phase	Primary course topics	Evidence produced
Initiation	Topic 1	Selection, business case, charter, kickoff
Planning	Topics 2 and 4	Requirements, baselines, risk responses
Execution	Topics 3, 5 and 6	Work, quality, engagement, resources
Monitoring & Controlling	Topics 3, 4, 5 and 7	Status, forecasts, issues, change, action
Closure	Topic 7	Acceptance, handover, lessons, benefits ownership

The seven-topic syllabus is preserved while the lifecycle makes the end-to-end operating system visible.

What Runs Across Every Phase

Integration

Keep decisions and artefacts consistent.

Stakeholders

Engage before decisions become resistance.

Risk

Refresh uncertainty as facts change.

Quality

Define, assure and control fitness for use.

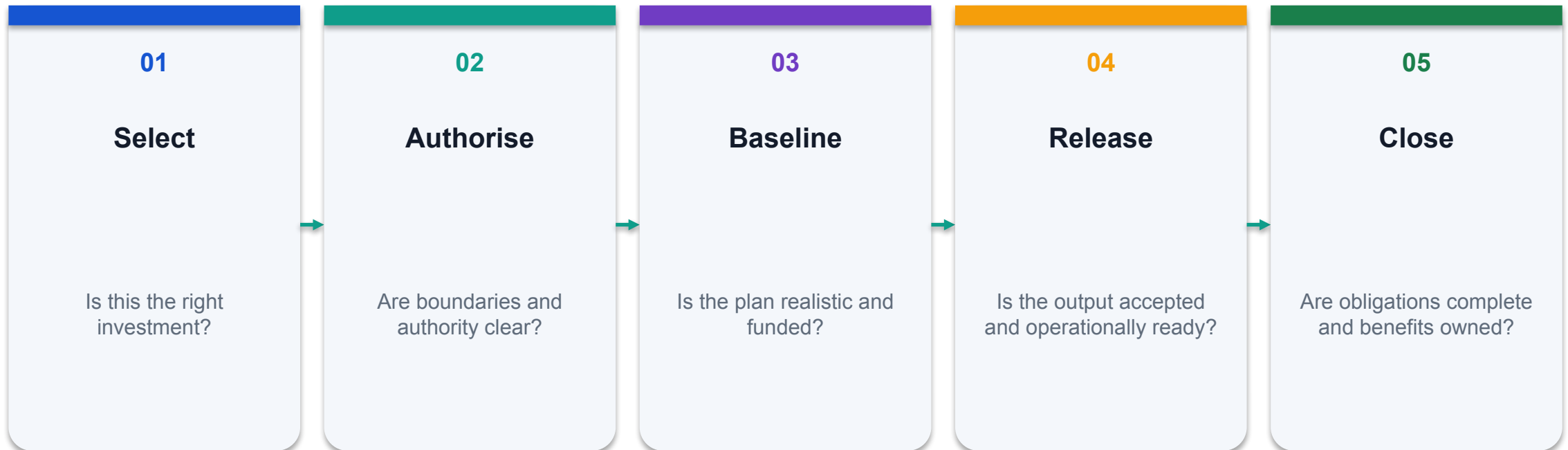
Resources

Match skills and capacity to priority work.

Governance

Escalate outside agreed tolerances.

Five Phase Gates for a Small Project



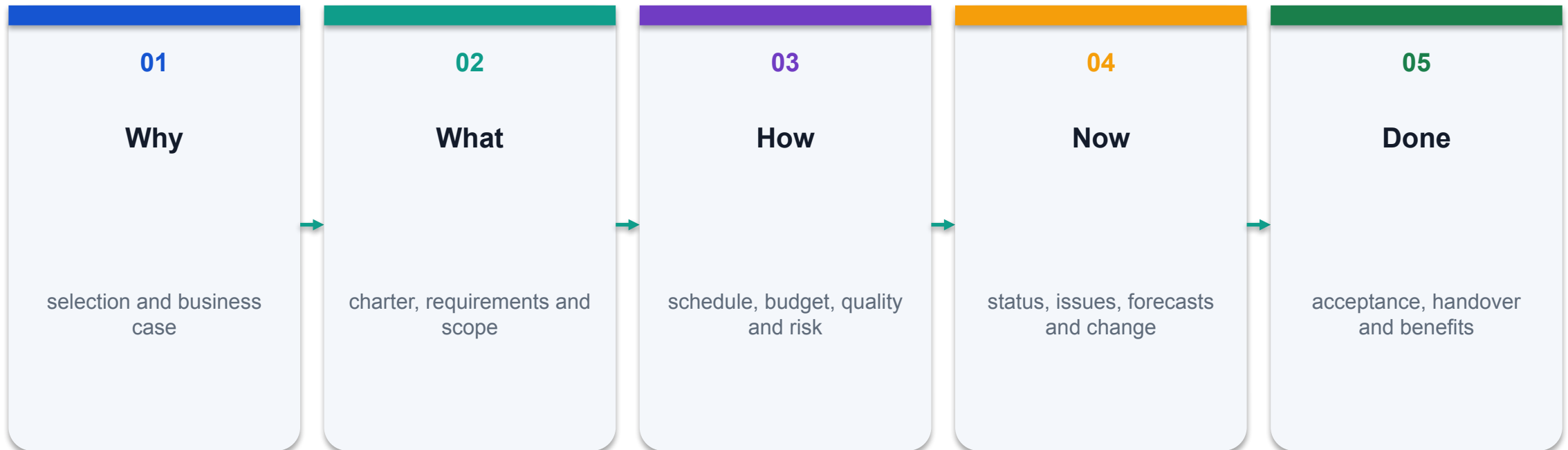
A gate is a decision with evidence and conditions—not a ceremonial meeting.

Tailoring Control to a Small Project

Need	Lean control	Escalate when
Scope	One charter, WBS and RTM	Success criteria or exclusions move
Schedule	Milestones plus critical path	Tolerance or launch date is threatened
Cost	Time-phased budget	Forecast exceeds authority
Risk	Prioritised register	Trigger, exposure or reserve changes materially
Change	One log and named approver	Trade-off crosses a tolerance

Tailoring removes low-value ceremony while preserving decision integrity.

The Connected Evidence Chain



Each artefact must feed the next decision; isolated templates are administration, not management.

Twelve Activities, One Project

Initiate

Activities 1–2

Plan

Activities 3–5 and 7

Execute

Activities 6, 8 and 9

Control

Activities 10–11

Close

Activity 12

TOPIC 01 · A1, K1 · LO1

Initiation of a Project

Small-project fit · value · charter · scope · team · kickoff

Small does not mean informal.

The smaller the project, the less waste it can absorb. A short charter, sharp boundaries and visible decision rights are the control system—not bureaucracy.

Initiation of a Project — Core Concepts I

01

A project is temporary, unique and value-seeking; operations are ongoing and repetitive.

02

A small project is normally under six months, has a limited budget and team, few dependencies and no significant strategic impact.

03

The project manager integrates people, decisions and trade-offs; they do not merely maintain a schedule.

04

Success balances value and outcomes with scope, schedule, cost, quality, resources and risk.

Initiation of a Project — Core Concepts II

05

Predictive, adaptive and hybrid approaches are choices driven by uncertainty, not labels of maturity.

06

The business case tests whether the project should exist; the charter authorises it to exist.

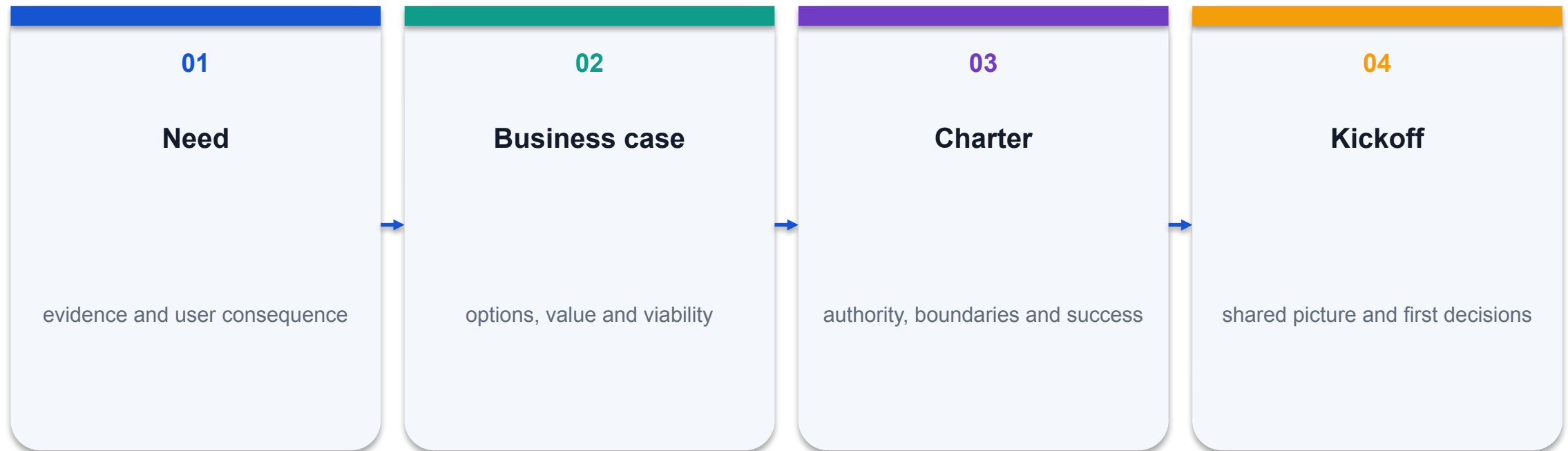
07

A useful charter names purpose, measurable success, scope boundaries, milestones, budget, major risks, team and sponsor authority.

08

A kickoff creates one shared picture of why, what, who, how and what happens next.

From Need to Authorised Work



Do not plan an unauthorised idea as though it were already a project.

Business Case vs Project Charter

Dimension	Business Case	Project Charter
Core question	Should we invest?	Are we authorised to proceed?
Focus	Options, benefits, costs, risk	Purpose, scope, team, milestones
Owner	Sponsor / business owner	Sponsor with project manager
Decision	Select or reject option	Authorize and delegate

Both are living decision records; neither is a ceremonial form.

Choosing a Development Approach

Signal	Predictive	Adaptive / Hybrid
Requirements	Stable and testable upfront	Evolving or discovery-heavy
Feedback	Available late or at gates	Available in short cycles
Change cost	High after commitment	Reduced through increments
Governance	Baseline and stage gates	Timeboxes plus control points

Approach follows uncertainty and reversibility—not fashion.

Project Work versus Operations

Dimension	Project	Operations
Purpose	Create a unique result	Sustain recurring value
Duration	Temporary	Ongoing
Change	Introduces change	Stabilises and improves work
Closure	Formal endpoint	Continues until redesigned

The same people may do both; the management system changes with the nature of the work.

The Small-Project Qualification Test

Time

Normally less than six months.

Budget

Limited enough for proportionate governance.

Team

Small core team with clear interfaces.

Dependencies

Few and manageable.

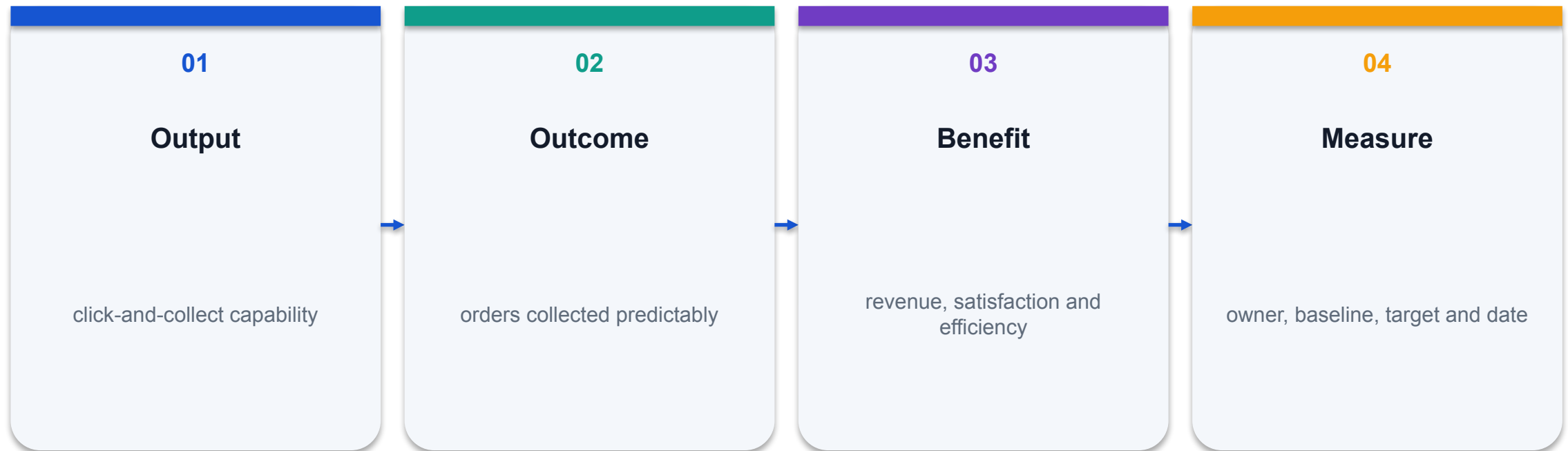
Impact

No significant strategic or enterprise-wide effect.

Uncertainty

Can be bounded through short learning cycles.

From Output to Benefits



Deliverables are produced by the project; benefits are usually realised with operations.

Project Selection Methods

Method	Best for	Watch for
Cost-benefit	Financial viability	Unsupported benefit assumptions
Weighted scoring	Multiple criteria	Biased weights
Payback	Liquidity and speed	Ignores later value
Mandatory driver	Compliance or safety	Treating obligation as optional

Use more than one lens when a small financial difference hides a major operational consequence.

Sponsor and Project Manager—Different Accountability

Sponsor

Owns investment, priority and escalation.

Project manager

Integrates delivery and advises decisions.

Business owner

Owns adoption and benefits after handover.

Functional manager

Provides capability and resolves resource conflicts.

Business Documents at Initiation

Artefact	Decision supported	Owner
Business case	Invest, hold or reject	Sponsor / business owner
Benefits plan	How value will be measured	Benefits owner
Project charter	Authorise and delegate	Sponsor with PM
Assumptions log	What must be validated	Project manager

Keep documents brief, but never combine ownership so completely that accountability disappears.

A Charter That Can Govern

Purpose

The problem and desired outcome.

Success

Measures, targets and dates.

Boundaries

Deliverables and exclusions.

Authority

Sponsor, PM and tolerances.

Frame

Milestones, budget and constraints.

Exposure

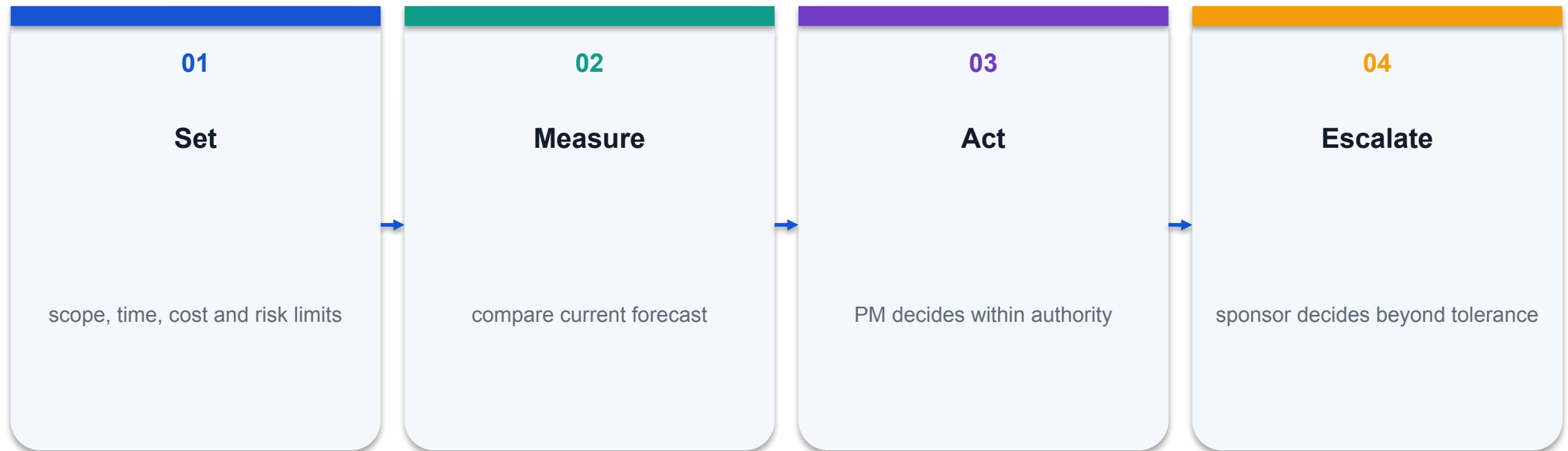
Assumptions, dependencies and high risks.

Assumption, Constraint, Dependency or Risk?

Type	Meaning	Example
Assumption	Believed true for planning	Outlet Wi-Fi capacity is sufficient
Constraint	Fixed limitation	S\$180,000 ceiling
Dependency	Reliance on other work	POS API access before integration
Risk	Uncertain event and impact	API access may be late

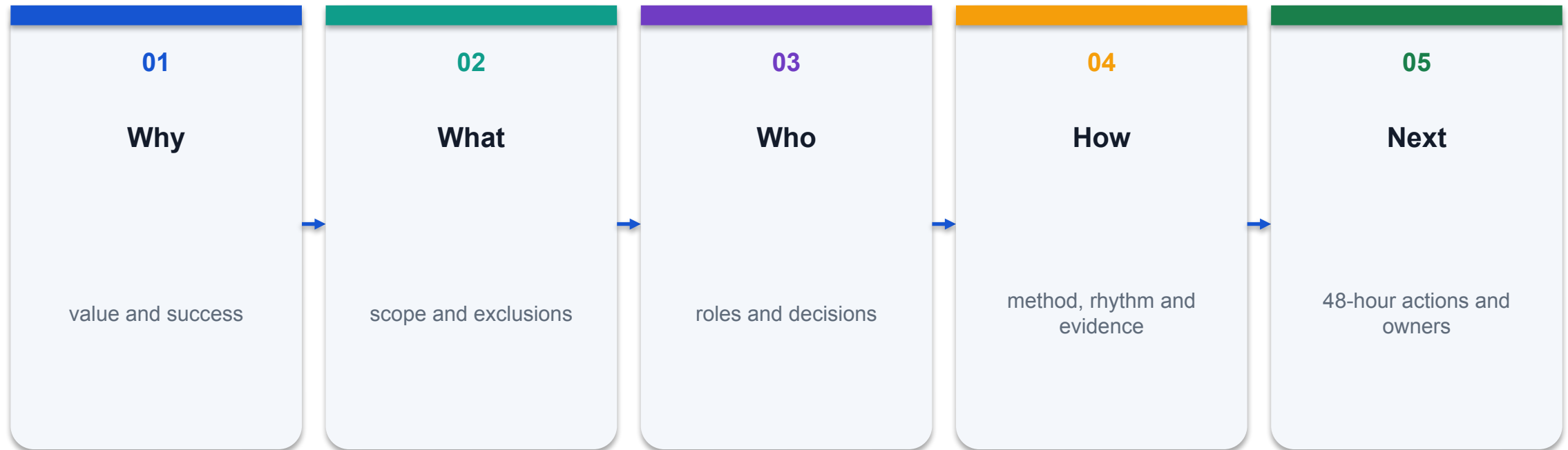
Classify correctly because each item needs a different response.

Lightweight Governance with Tolerances



Authority without tolerances creates hesitation; tolerances without reporting create surprise.

Kickoff as an Alignment Decision



A good kickoff reveals mismatched assumptions early enough to repair them cheaply.

Activity 1 Tool Pack

XLSX

02 Cost Benefit Analysis

Editable evidence file

XLSX

01 Small Project Screening and Selection

Editable evidence file

DECISION ENABLED

A completed small-project screen, weighted option scorecard, cost-benefit check and sponsor-ready selection recommendation.

Screen and Select the HarbourHub Pilot

THE SITUATION

HarbourHub Fresh has three competing improvement ideas: click-and-collect, a loyalty app, and a warehouse automation study. Only one can be funded this quarter. The sponsor needs a transparent selection decision and evidence that the selected work fits the small-project profile.

WHAT YOU'LL DO

Apply the small-project screen, compare options using weighted criteria, test the financial case and document the recommendation, assumptions and rejection reasons.

YOU'LL PRODUCE

A completed small-project screen, weighted option scorecard, cost-benefit check and sponsor-ready selection recommendation.

DONE WHEN

The selected option satisfies every small-project criterion, the weighted score reconciles, the financial downside is visible and the recommendation states a clear gate decision.

Tool: Small Project Screening, Multi-Criteria Selection and Cost-Benefit Analysis

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 1 folder.

Debrief — Activity 1: What It Proves

1

Selection prevents the team from efficiently delivering the wrong project.

2

Weights expose priorities; narrative rationale prevents false precision.

3

A selected idea is still not authorised work until the charter gate is passed.

Activity 2 Tool Pack

DOCX

05 Business Case on a Page

Editable evidence file

XLSX

08 Project Charter Template

Editable evidence file

DECISION ENABLED

A one-page business case, signed-ready charter, assumptions and constraints log, milestone frame, decision tolerances and kickoff action list.

Authorise the Pilot and Run the Kickoff Gate

THE SITUATION

The click-and-collect pilot has been selected, but Operations, Finance and the two vendors hold different assumptions about success, exclusions and decision authority. The sponsor will authorise delivery only after a concise gate review.

WHAT YOU'LL DO

Convert the selected option into a measurable business case and charter, define boundaries, name decision rights and conduct a short sponsor gate and kickoff.

YOU'LL PRODUCE

A one-page business case, signed-ready charter, assumptions and constraints log, milestone frame, decision tolerances and kickoff action list.

DONE WHEN

The charter names one accountable sponsor, measurable success, explicit exclusions, decision tolerances and an unambiguous authorisation decision within the small-project constraints.

Tool: Business Case on a Page and Project Charter

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 2 folder.

Debrief — Activity 2: What It Proves

1

The business case justifies investment; the charter grants authority.

2

Decision tolerances make lightweight governance possible.

3

Kickoff converts signed paper into shared commitments and immediate action.

Topic 01 Recap

Decision

A project is temporary, unique and value-seeking; operations are ongoing and repetitive.

Method

Success balances value and outcomes with scope, schedule, cost, quality, resources and risk.

Evidence

A one-page business case, signed-ready charter, assumptions and constraints log, milestone frame, decision tolerances and kickoff action list.

Capability

Define scope, success, authority and team arrangements for an executable small-project start (A1, K1).

TOPIC 02 · A2, K2 · LO2

Develop a Project Plan

Requirements · scope baseline · WBS · network · schedule · budget

A plan is a connected model, not a collection of files.

Requirements, WBS, dependencies, dates, resources and money must describe the same work. If one changes, the others must move with it.

Develop a Project Plan — Core Concepts I

01

The plan integrates subsidiary decisions; separate scope, schedule and cost files are not yet a project plan.

02

Progressive elaboration and rolling-wave planning add detail as uncertainty reduces without abandoning control.

03

Requirements must be testable, prioritised and traceable from need through deliverable and acceptance evidence.

04

The scope statement defines deliverables, exclusions, constraints, assumptions and acceptance criteria.

Develop a Project Plan — Core Concepts II

05

The WBS decomposes deliverables to manageable work packages; the WBS dictionary removes ambiguity.

06

Dependencies create the network; the longest path sets the earliest finish and carries zero or least float.

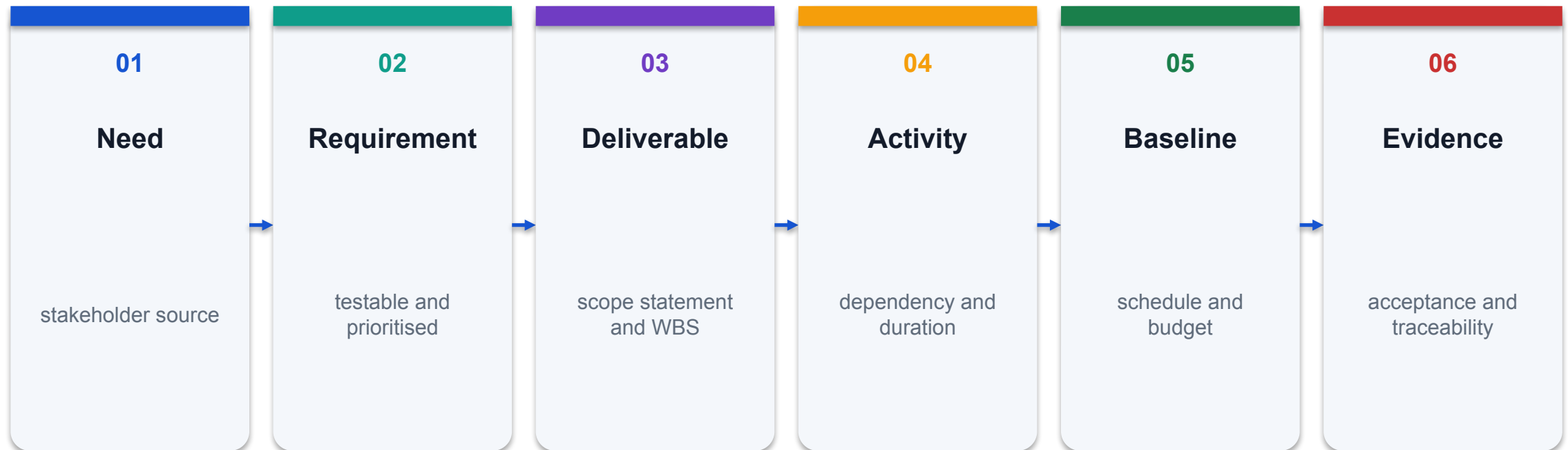
07

Duration estimates should separate work, waiting, risk and resource availability; use analogous, parametric, bottom-up or three-point methods deliberately.

08

The cost baseline is time-phased; contingency reserve covers known-unknowns while management reserve covers unknown-unknowns.

The Planning Chain



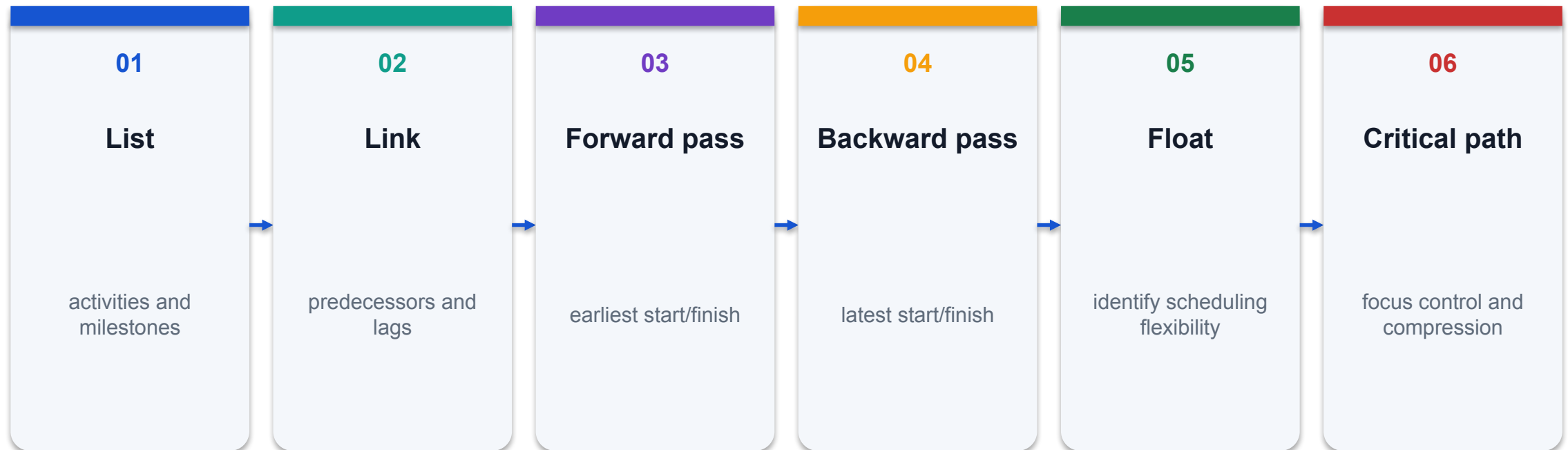
A broken link creates either orphan scope or unplanned work.

Estimating Methods

Method	Best use	Main risk
Analogous	Early estimate from similar work	False similarity
Parametric	Repeatable unit rates	Bad rate or scale
Bottom-up	Defined work packages	Time and detail cost
Three-point	Uncertain durations/costs	Optimism in inputs

Record the basis of estimate so future variance can be explained.

Critical Path Logic



Recalculate after every approved change—the critical path can move.

The Integrated Project Plan

Scope

What is authorised and accepted.

Schedule

Sequence, duration and milestones.

Cost

Time-phased funding and reserves.

Quality

Measures, assurance and control.

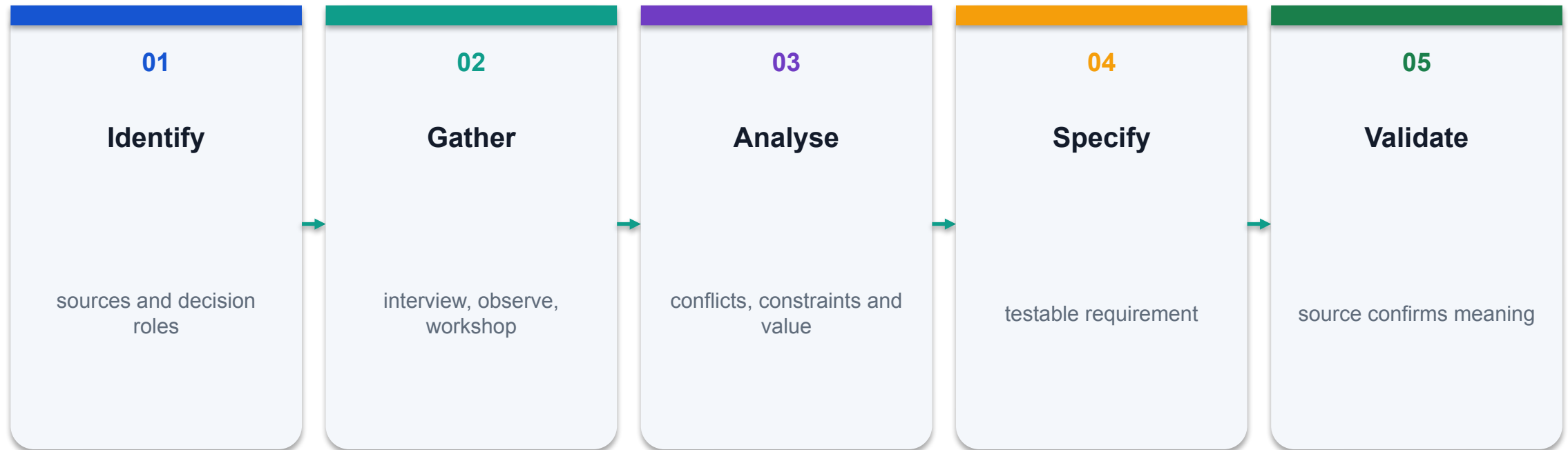
Resources

Skills, capacity and procurement.

Risk & stakeholders

Responses, engagement and governance.

Requirements Elicitation



Elicitation is discovery and negotiation—not taking an order from one stakeholder.

Weak versus Testable Requirements

Weak wording	Testable wording	Evidence
Fast collection	90% ready within 10 minutes	Timestamp sample
Accurate picking	At least 98% line accuracy	Check sheet
Easy to use	New staff completes task after 30-minute training	Observed test
Reliable	99.5% availability in pilot hours	System log

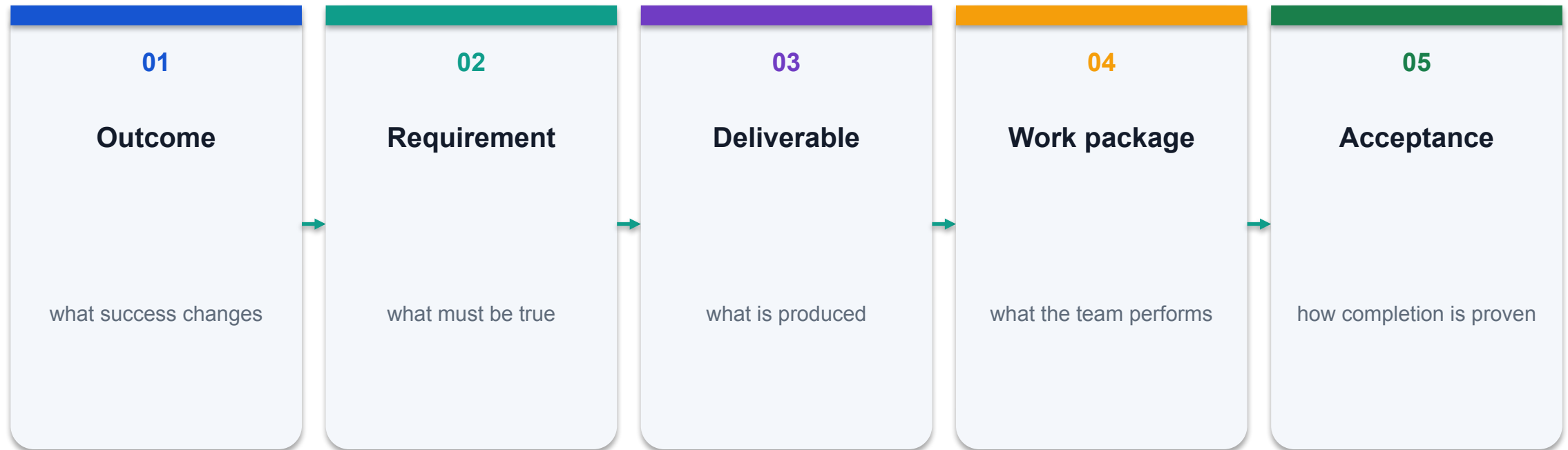
A measure without a method or status date is still ambiguous.

Prioritisation Choices

Method	Decision logic	Use when
MoSCoW	Must / Should / Could / Won't	Time-boxed scope
Weighted scoring	Value across criteria	Competing options
Kano	Basic, performance, delight	Customer experience
Trade-off sliders	Relative constraint priority	Governance alignment

Priorities must remain traceable to outcomes, not just stakeholder preference.

Traceability from Need to Evidence



Trace both directions: every need is delivered, and every deliverable has a reason.

Scope Statement Essentials

Product scope

Characteristics of the result.

Project scope

Work required to create it.

Exclusions

What is deliberately not included.

Acceptance

Who approves against which criteria.

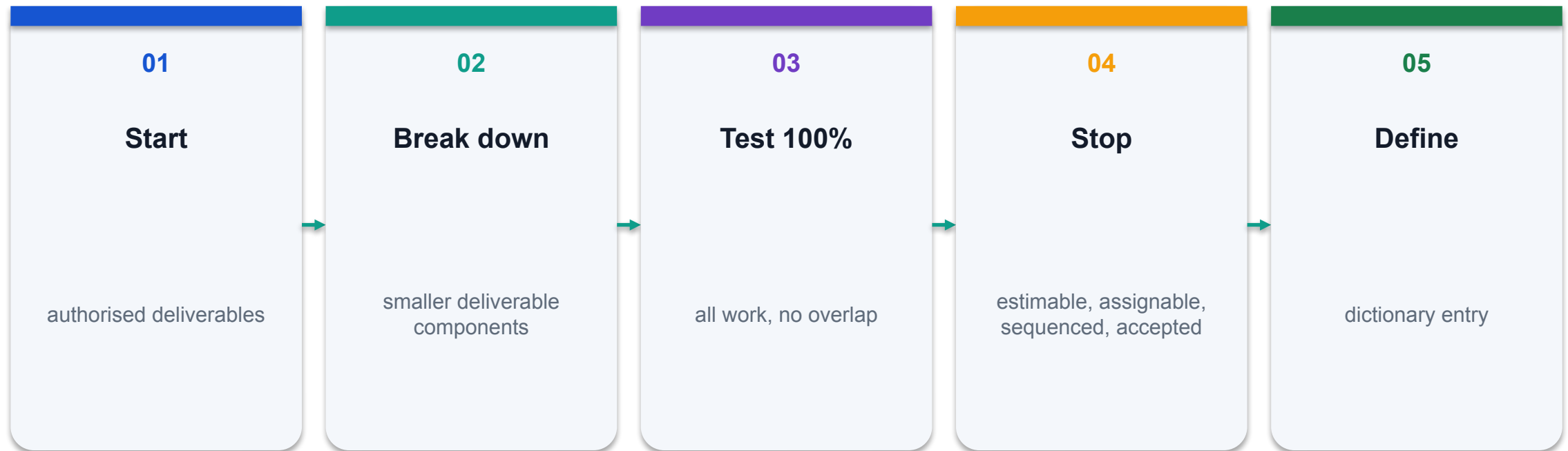
Constraints

Limits the plan must respect.

Assumptions

Planning beliefs to validate.

WBS Decomposition



Do not decompose by department when the deliverable crosses departments.

WBS Dictionary—Why It Matters

Field	Decision enabled	Failure prevented
Description	Shared meaning	Different estimates for different work
Owner	Accountability	Unclaimed delivery
Acceptance	Definition of complete	Almost done forever
Dependencies	Sequence	Late discovery
Estimate basis	Confidence and revision	Unexplained padding

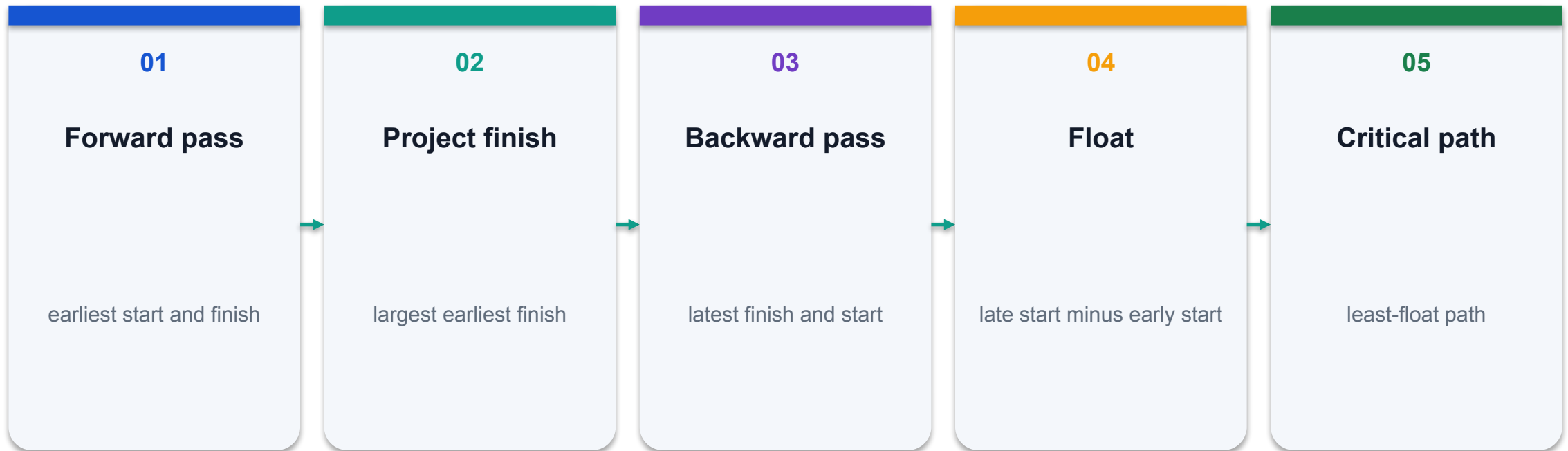
The diagram shows structure; the dictionary makes the structure manageable.

Dependency Relationships

Type	Logic	Example
Finish-to-start	A finishes before B starts	Configure before integration test
Start-to-start	A starts before B starts	Draft training after testing begins
Finish-to-finish	A finishes before B finishes	Documentation completes with testing
Start-to-finish	Rare transition logic	Old support ends after new starts

Leads and lags change timing; they do not remove the underlying dependency.

Critical Path Calculation



The critical path is the longest-duration route through current logic, not the most important-looking task.

Estimating Methods

Method	Strength	Limitation
Analogous	Fast early estimate	Lower detail
Parametric	Repeatable unit logic	Needs valid relationship
Bottom-up	Detailed and accountable	Time-consuming
Three-point	Makes uncertainty visible	Still depends on judgement

Record the basis so the estimate can be challenged and updated without starting over.

Cost Baseline and Reserves

Work-package estimates

Rolled up from authorised scope.

Contingency reserve

Funded responses for identified risk.

Cost baseline

Time-phased performance reference.

Management reserve

Held outside baseline for unknown-unknowns.

Project budget

Cost baseline plus management reserve.

Activity 3 Tool Pack

XLSX

25 Requirements Traceability Matrix

Editable evidence file

XLSX

03 Requirements Workshop and Stakeholder Register

Editable evidence file

DECISION ENABLED

A stakeholder register, workshop decision log and requirements traceability matrix with priorities, owners, deliverables and measurable acceptance evidence.

Discover Requirements and Map Acceptance Evidence

THE SITUATION

Outlet supervisors want speed, Finance wants payment control, customers want predictable collection, and the POS vendor wants a stable interface. The charter is approved, but the stated needs are not testable.

WHAT YOU'LL DO

Identify sources, facilitate a structured workshop, convert needs into testable requirements, prioritise with MoSCoW and define acceptance evidence and ownership.

YOU'LL PRODUCE

A stakeholder register, workshop decision log and requirements traceability matrix with priorities, owners, deliverables and measurable acceptance evidence.

DONE WHEN

Every prioritised requirement has a source, rationale, owner, deliverable and testable acceptance method; unresolved conflicts have named decision owners and dates.

Tool: Requirements Workshop, Stakeholder Register and Traceability Matrix

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 3 folder.

Debrief — Activity 3: What It Proves

1

A requirement is not complete until acceptance can be observed or measured.

2

Prioritisation is a trade-off decision, not a label applied by the loudest stakeholder.

3

Traceability protects value when scope changes later.

Activity 4 Tool Pack

PPTX

24 Work Breakdown Structure

Editable evidence file

XLSX

24 WBS Dictionary

Editable evidence file

XLSX

25 Requirements Traceability Matrix

Editable evidence file

DECISION ENABLED

A scope statement, WBS, WBS dictionary and repaired traceability matrix with no missing or duplicated work.

Build the Scope Baseline with WBS and Dictionary

THE SITUATION

The team has 14 prioritised requirements, but the vendor proposal groups work by department and omits outlet readiness, training and handover. Estimates cannot be trusted until the total scope is visible.

WHAT YOU'LL DO

Create a deliverable-oriented WBS, apply the 100% rule, define work packages in the dictionary and reconcile every requirement and acceptance criterion to scope.

YOU'LL PRODUCE

A scope statement, WBS, WBS dictionary and repaired traceability matrix with no missing or duplicated work.

DONE WHEN

The WBS covers 100% of authorised work without overlap, each work package is manageable and every requirement traces to acceptance-ready scope.

Tool: Work Breakdown Structure, WBS Dictionary and Requirements Traceability Matrix

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 4 folder.

Debrief — Activity 4: What It Proves

1

The WBS describes all project work; the schedule describes when that work occurs.

2

Transition and project-management work are real scope even when customers never see them.

3

A strong dictionary prevents different teams from estimating different meanings.

Activity 5 Tool Pack

XLSX

28 Schedule Network Diagram

Editable evidence file

XLSX

29 - Gantt Chart - Schedule - Daily Weekly Monthly and Phases

Editable evidence file

XLSX

32 Project Budget - Planned vs Actual - Weekly and Monthly

Editable evidence file

DECISION ENABLED

A network with forward/backward passes, critical path, Gantt milestones, basis of estimates and a time-phased cost baseline plus contingency within the authorised ceiling.

Develop the Integrated Schedule and Cost Baseline

THE SITUATION

The launch must precede Lunar New Year trading. The vendor has proposed a date without dependency logic, resource calendars or reserves. Finance requires a time-phased baseline within S\$180,000.

WHAT YOU'LL DO

Sequence work packages, calculate the critical path and float, estimate duration and cost using explicit methods, set milestones and reconcile schedule, budget and reserve.

YOU'LL PRODUCE

A network with forward/backward passes, critical path, Gantt milestones, basis of estimates and a time-phased cost baseline plus contingency within the authorised ceiling.

DONE WHEN

The dependency network calculates correctly, the plan fits the five-month limit, all authorised work is time-phased and funded, and baseline plus contingency stays within S\$180,000.

Tool: Network Diagram, Gantt Schedule and Budget versus Actual

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 5 folder.

Debrief — Activity 5: What It Proves

1

A Gantt is the presentation of schedule logic, not the logic itself.

2

The critical path can change after every update or approved change.

3

Baseline and forecast must remain separate so variance can be explained.

Topic 02 Recap

Decision

The plan integrates subsidiary decisions; separate scope, schedule and cost files are not yet a project plan.

Method

The scope statement defines deliverables, exclusions, constraints, assumptions and acceptance criteria.

Evidence

A network with forward/backward passes, critical path, Gantt milestones, basis of estimates and a time-phased cost baseline plus contingency within the authorised ceiling.

Capability

Sequence, estimate and fund the authorised scope as one realistic plan (A2, K2).

TOPIC 03 · A3, K3 · LO3

Apply Methods and Tools to Drive a Project

Quality planning · flow · root cause · Pareto · improvement

Use the lightest tool that changes the next decision.

A tool earns its place when it turns uncertain work into visible evidence: flow, cause, variation, priority or ownership.

Apply Methods and Tools to Drive a Project — Core Concepts I

01

A method defines how work is managed; a tool makes information visible. Choose both for the decision needed.

02

Predictive plans optimise sequence and baseline control; adaptive boards optimise flow and feedback; hybrid combines both intentionally.

03

Quality means conformance to requirements and fitness for use, not luxury or perfection.

04

Prevention and appraisal are usually cheaper than internal and external failure.

Apply Methods and Tools to Drive a Project — Core Concepts II

05

A check sheet captures facts, a control chart detects process behaviour, Pareto prioritises causes, and fishbone plus 5 Whys investigates root cause.

06

Root-cause analysis must end at a process or system cause the team can influence, not blame a person.

07

Visual work management exposes blocked work and excessive WIP before the schedule report does.

08

Continuous improvement closes the loop through a named action, owner, measure and review date.

Select the Tool for the Question

Question	Best starting tool	Output
What happened?	Check sheet	Reliable facts
Why might it happen?	Fishbone	Cause families
What is the root?	5 Whys	Actionable chain
Which causes matter most?	Pareto	Vital few
Is the process stable?	Control chart	Common vs special cause

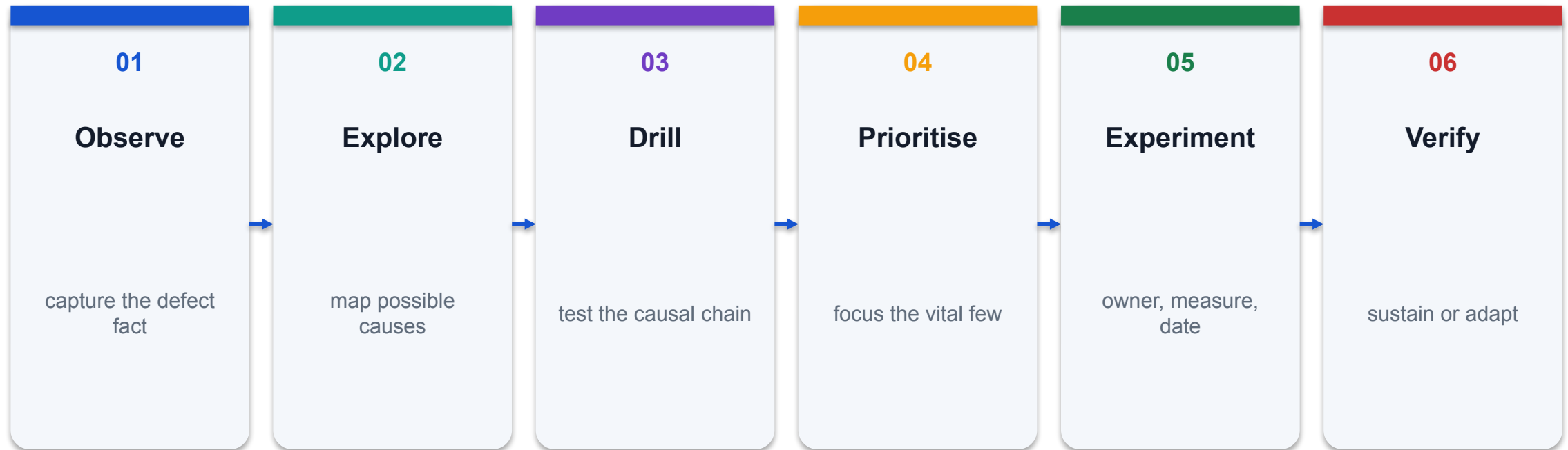
Tools work as a sequence: facts → causes → priority → behaviour → action.

Cost of Quality

Category	Purpose	Example
Prevention	Stop defects forming	Training and standards
Appraisal	Detect before release	Review and testing
Internal failure	Fix before customer	Rework and retest
External failure	Recover after release	Returns and reputation

Invest upstream where a dollar of prevention avoids many dollars of failure.

Root Cause to Improvement



Stopping at a cause without a verified action is analysis theatre.

Method versus Tool

Element	Purpose	Example
Method	Defines how work is managed	Predictive, Scrum, Kanban, PDCA
Tool	Makes evidence visible	WBS, board, chart, log
Policy	Defines operating rule	WIP limit, acceptance rule
Cadence	Defines review rhythm	Daily review, weekly status

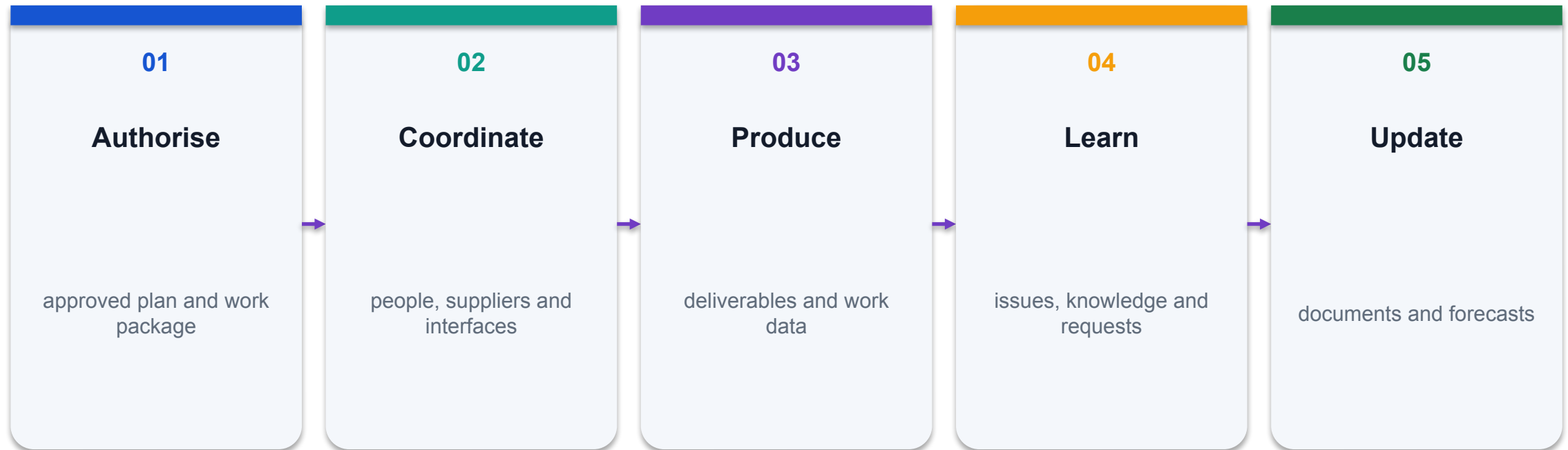
A familiar tool used without a decision rule becomes decoration.

Predictive, Adaptive and Hybrid

Signal	Predictive	Adaptive / hybrid
Requirements	Stable and knowable	Evolving or discovery-heavy
Delivery	Larger gated releases	Short usable increments
Change	Controlled against baseline	Expected inside prioritised backlog
Best fit	Low uncertainty, costly rework	Fast feedback, reversible choices

Tailor by deliverable when different parts of one small project carry different uncertainty.

Direct and Manage Project Work



Execution produces both deliverables and information needed for control.

A Lean PMIS for a Small Project

Source of truth

Named location and owner.

Version control

Number, date, author and status.

Decision log

What, who, when and rationale.

Access

Right people, current artefact.

Archive

Superseded records remain traceable.

Security

Sensitive content has proportionate access.

Quality Planning, Assurance and Control

Discipline	Question	Evidence
Planning	What quality is required?	Metrics, methods and acceptance
Assurance	Will the process achieve it?	Review, audit, coaching
Control	Does the result conform?	Inspection, tests and charts
Improvement	How will the system change?	Experiment and verification

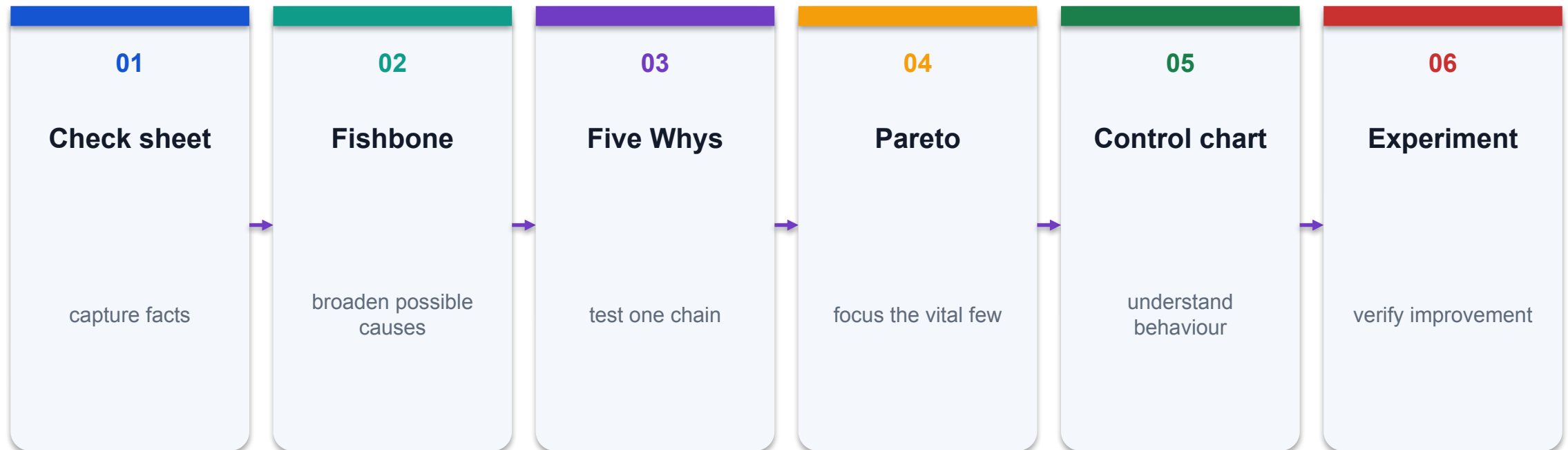
Inspection finds defects; assurance and improvement reduce the chance of creating them.

HarbourHub Quality-Cost Decision

Option	Near-term effect	Decision signal
Add catalogue-data validation	S\$4,000 prevention spend	Reduces rejection and re-pick risk
Test only at pilot launch	Lower spend now	Defects discovered with customers waiting
Accept manual correction	No tooling change	Recurring labour and queue growth
Decision	Fund validation before rollout	Cheaper than repeated external failure

The team chooses prevention because one avoided launch-week failure can repay the control.

Quality Tool Sequence



Do not jump from anecdote directly to solution.

Common Cause versus Special Cause

Signal	Meaning	Response
Stable random variation	System behaviour	Improve the process
Point beyond limits	Potential special cause	Investigate event
Run or trend	Non-random pattern	Test changed condition
Within limits but off specification	Stable but incapable	Redesign process

Control limits come from process data; specification limits come from requirements.

Kanban Operating Policies

Visualise

Show status, owner and blockers.

Limit WIP

Stop starting; start finishing.

Manage flow

Watch ageing and bottlenecks.

Make policies explicit

Entry, exit and priority rules.

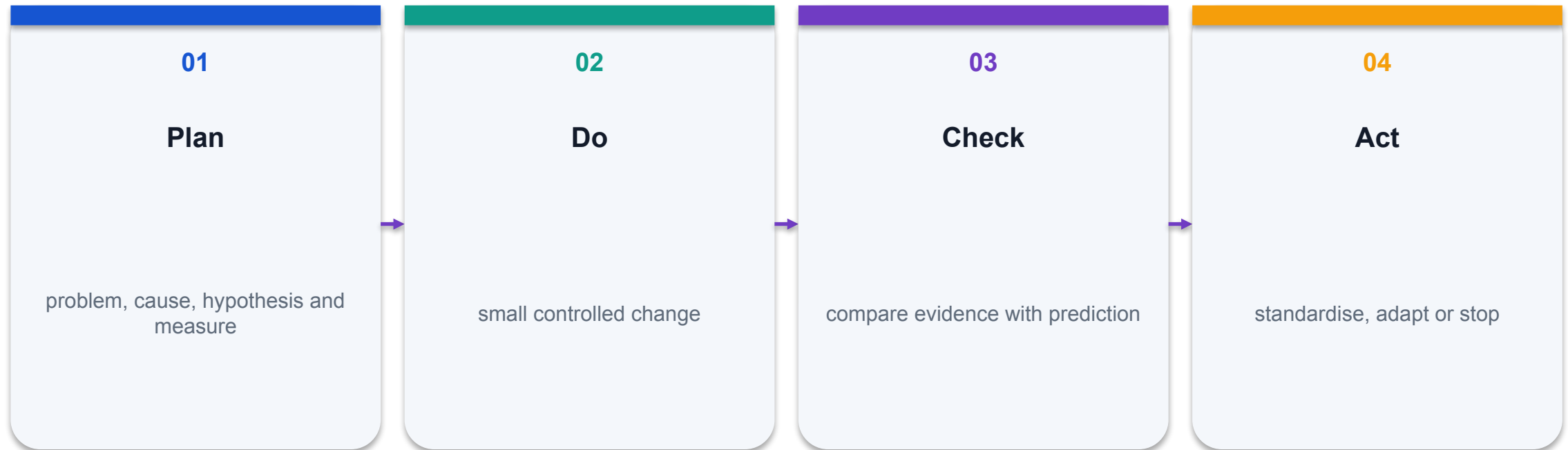
Feedback

Review at a useful cadence.

Improve

Run small measurable experiments.

PDCA Improvement Loop



Improvement is complete only when the new practice is verified and sustained.

Manage Project Knowledge

Explicit

Documents, data and procedures.

Tacit

Experience and judgement shared through work.

Create

Reviews, experiments and problem solving.

Use

Decision support during delivery.

Transfer

Handover and lessons for future teams.

Activity 6 Tool Pack

30 Kanban Board

XLSX · editable project tool

35 Check Sheets

XLSX · editable project tool

34 Control Chart

XLSX · editable project tool

06 Delivery Approach and Quality Plan

XLSX · editable project tool

Tailor the Delivery Method and Quality Plan

THE SITUATION

The catalogue and POS interface need predictable milestone control, while the outlet workflow needs rapid testing and feedback. The team must choose a hybrid delivery rhythm and define quality before build begins.

WHAT YOU'LL DO

Assess uncertainty and reversibility, tailor predictive and adaptive controls, define the Definition of Done, quality metrics, review cadence, WIP limits and evidence rules.

YOU'LL PRODUCE

A documented hybrid approach, phase and iteration rhythm, quality management plan, Definition of Done, Kanban policy and measurement plan.

DONE WHEN

The approach matches uncertainty, quality is measurable before work begins, WIP and evidence rules are explicit, and predictive/adaptive controls do not contradict one another.

Tool: Delivery Approach Canvas, Quality Plan and Kanban Board

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 6 folder.

Debrief — Activity 6: What It Proves

1

Hybrid is intentional tailoring, not running two incompatible systems in parallel.

2

Quality must be designed into the workflow before it is inspected at the end.

3

A tool earns its place only when it changes a decision or behaviour.

Activity 10 Tool Pack

35 Check Sheets

XLSX · editable project tool

37 Ishikawa Diagram

XLSX · editable project tool

38 Five Whys

XLSX · editable project tool

36 Pareto Principle Template

XLSX · editable project tool

34 Control Chart

XLSX · editable project tool

Diagnose the Pilot Quality Failure

THE SITUATION

During testing, only 82% of orders are ready on time and pick accuracy is 94%. The vendor blames store staff; store staff blame inventory updates. The sponsor wants action before facts and hypotheses are separated.

WHAT YOU'LL DO

Reconcile defect data, broaden possible causes, test a causal chain, prioritise the vital few, interpret process behaviour and design one measurable corrective experiment.

YOU'LL PRODUCE

A check sheet, fishbone, five-whys chain, Pareto chart, control-chart interpretation and SMART corrective experiment.

DONE WHEN

Data totals reconcile, Pareto reaches 100%, the causal chain is actionable, the control interpretation matches the signal and the experiment has an owner and verification date.

Tool: Check Sheet, Fishbone, Five Whys, Pareto and Control Chart

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 10 folder.

Debrief — Activity 10: What It Proves

1

Fishbone broadens, Five Whys deepens and Pareto prioritises.

2

Control limits describe process behaviour; specifications describe customer need.

3

Improvement is project work only when funded, owned and verified.

Topic 03 Recap

Decision

A method defines how work is managed; a tool makes information visible. Choose both for the decision needed.

Method

Prevention and appraisal are usually cheaper than internal and external failure.

Evidence

A check sheet, fishbone, five-whys chain, Pareto chart, control-chart interpretation and SMART corrective experiment.

Capability

Use evidence and root-cause tools to select a controlled corrective experiment (A3, K3).

TOPIC 04 · A4, K4 · LO4

Manage Project Risk

Identify · assess · respond · contingency · monitor

Risk management buys options before they become expensive.

The register matters only when it changes funding, sequencing, ownership or contingency before a trigger becomes an issue.

Manage Project Risk — Core Concepts I

01

Risk is an uncertain event or condition with a positive or negative effect; an issue has already happened.

02

Write risks as cause-event-impact statements so the response targets the cause or reduces the impact.

03

The risk register records owner, probability, impact, exposure, triggers, response, contingency and residual risk.

04

Qualitative analysis prioritises quickly; quantitative analysis estimates the range of outcomes and decision value.

Manage Project Risk — Core Concepts II

05

Threat responses are avoid, mitigate, transfer, accept or escalate; opportunity responses are exploit, enhance, share, accept or escalate.

06

A contingency plan is triggered by an observable threshold and funded from contingency reserve.

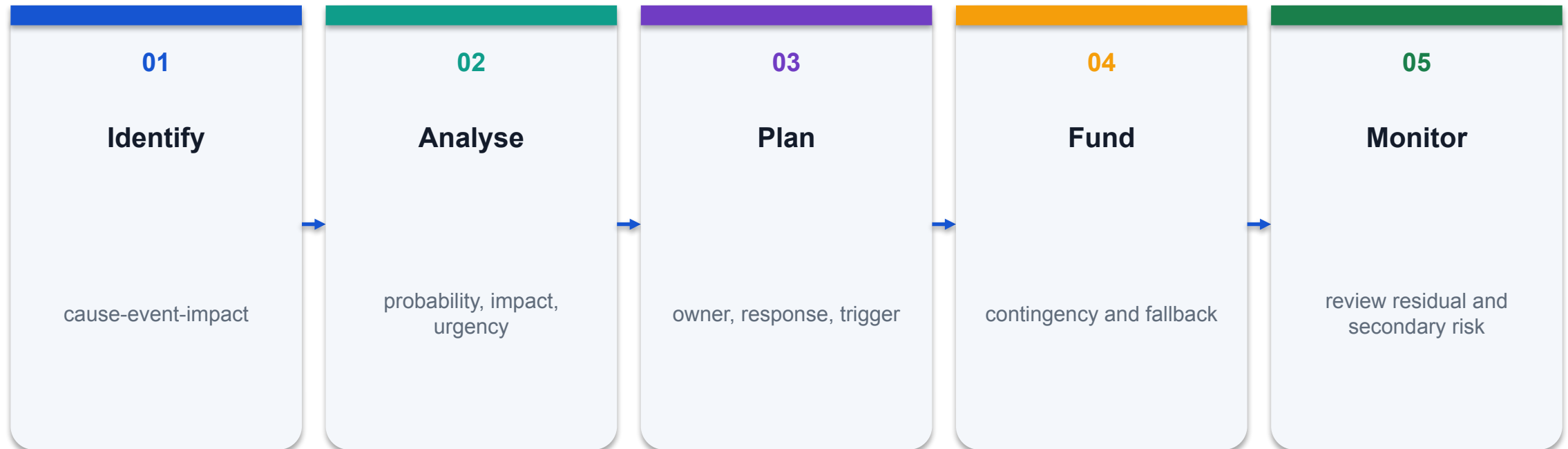
07

Decision trees compare expected monetary value; tornado charts reveal which assumptions deserve the most attention.

08

Risk reviews are recurring decisions, not a register created once and archived.

The Risk Cycle



Triggered risk becomes an issue and moves into action management.

Threat Response Choices

Response	What changes	Example
Avoid	Remove the threat source	Change approach
Mitigate	Reduce probability or impact	Prototype early
Transfer	Shift financial consequence	Insurance / contract
Accept	Monitor and prepare	Contingency plan
Escalate	Move beyond authority	Sponsor decision

Acceptance is a decision with monitoring—not forgetting the risk.

Qualitative vs Quantitative Analysis

Lens	Qualitative	Quantitative
Speed	Fast prioritisation	More data and effort
Output	Relative exposure	Range, EMV or sensitivity
Use	Most project risks	Material decision uncertainty
Caution	Scoring bias	False precision

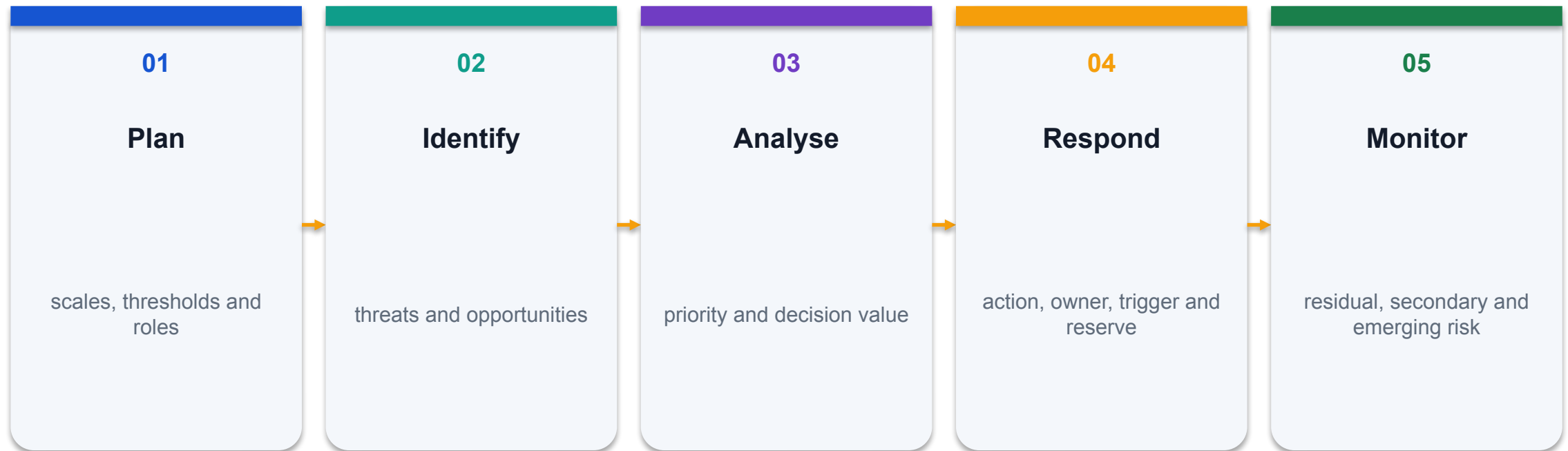
Use quantitative analysis only where the decision value justifies the work.

Risk, Issue and Assumption

Item	Time orientation	Management action
Risk	May happen	Owner, trigger and response
Issue	Has happened	Contain, resolve and escalate
Assumption	Believed true	Validate by a date
Constraint	Known limit	Plan within or change formally

A triggered risk becomes an issue; its response owner moves from readiness to action.

The Risk Management Cycle



Risk management repeats whenever scope, schedule, suppliers or the environment change.

Risk Breakdown Structure

Technical

Design, interface, data and quality.

External

Regulation, market, weather and supplier.

Organisational

Priority, funding, skills and governance.

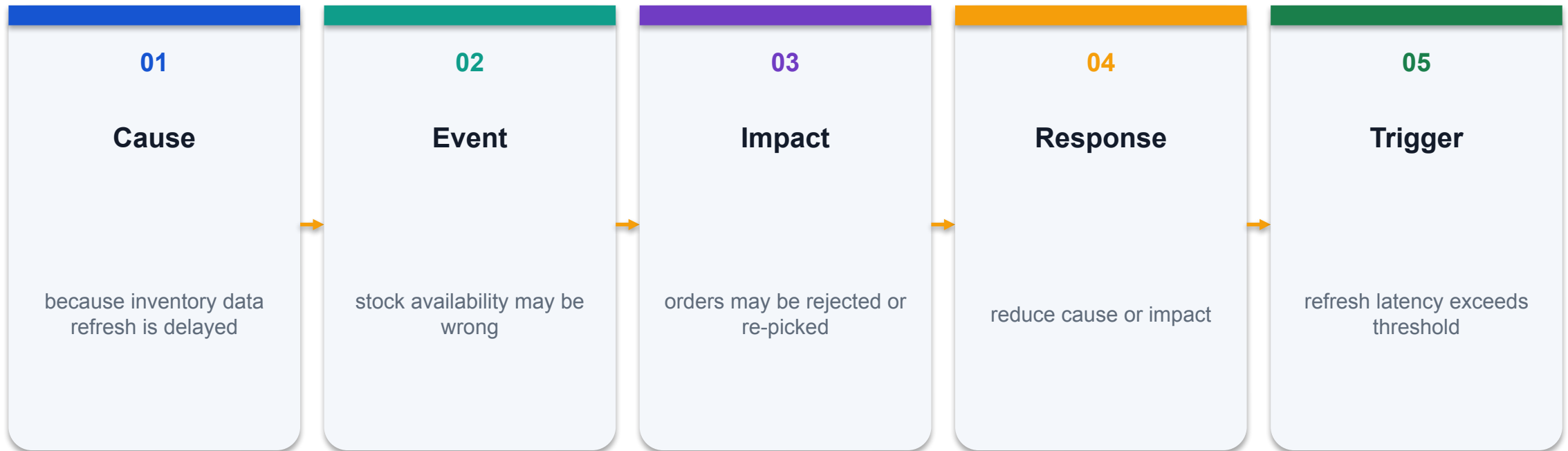
Project management

Estimate, dependency, communication and scope.

Opportunity

Faster adoption, reuse, cost saving and capability.

Writing an Actionable Risk



Cause-event-impact wording prevents vague risks such as 'vendor risk'.

Qualitative Risk Factors

Factor	Question	Use
Probability	How likely?	Exposure priority
Impact	How severe?	Tolerance and reserve
Urgency	How soon must we act?	Response timing
Proximity	How soon may it occur?	Monitoring cadence
Dormancy	How long before impact appears?	Early detection

Two risks with the same score may require different action timing.

HarbourHub Threat-Response Workshop

Named threat	Chosen response	Owned action and trigger
Inventory feed is late	Mitigate	IT lead adds latency alert; trigger >15 minutes
Vendor misses integration date	Transfer + mitigate	Milestone clause plus weekly interface test
Outlet rejects new process	Mitigate	Ops lead pilots job aid and observes first 20 orders
Cyber approval exceeds tolerance	Escalate	Sponsor decides scope trade by gate date
Residual launch disruption	Accept actively	Fund two-day floor support contingency

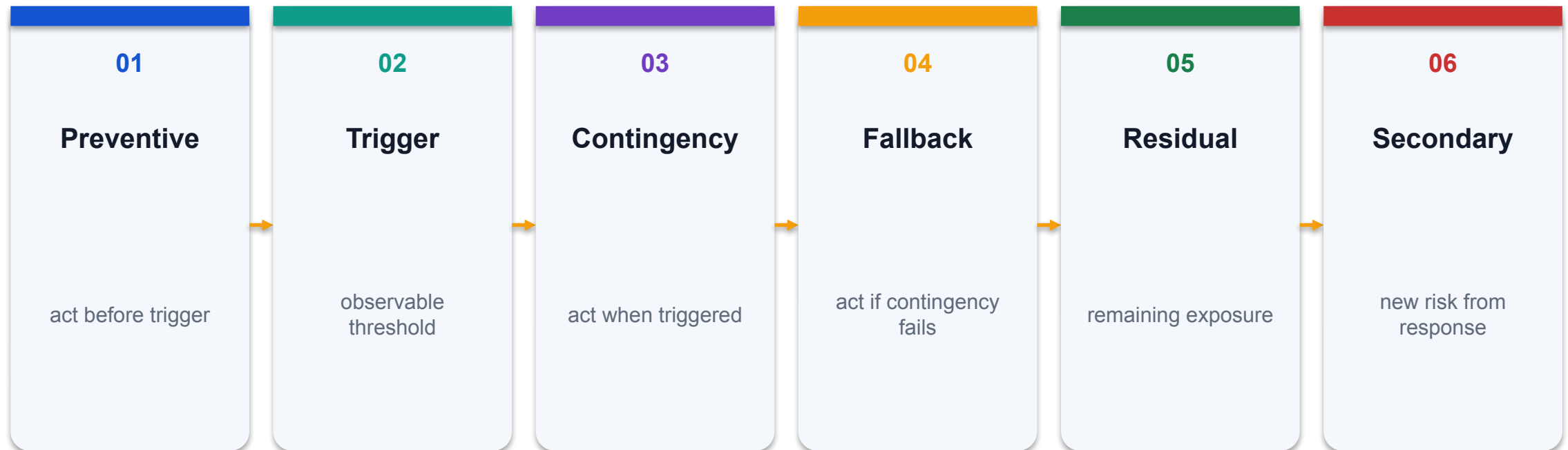
Each response is specific enough to schedule, fund, monitor and verify.

Opportunity Response Strategies

Strategy	Intent	Example
Exploit	Ensure opportunity occurs	Assign best resources
Enhance	Increase probability or benefit	Accelerate pilot learning
Share	Partner for capability	Revenue-sharing supplier
Accept	Take benefit if it occurs	No proactive spend
Escalate	Move to programme or strategy	Enterprise reuse

Positive uncertainty deserves ownership and deliberate action too.

Response Architecture



A response is not operational until timing, ownership and funding are explicit.

Decision Tree and Sensitivity

Tool	Question answered	Caution
Decision tree	Which option has better expected value?	Average value can hide tail consequences
Tornado chart	Which assumption drives outcome most?	Range quality determines insight
Probability-impact matrix	What needs attention first?	Ordinal scores are not precise money

Use more analysis only when the decision value exceeds the analysis effort.

Risk Review Agenda

Triggered

Promote to issue and execute response.

Changed

Re-score probability, impact and timing.

New

Add emerging and secondary risk.

Obsolete

Close with rationale.

Reserve

Reconcile use and remaining exposure.

Escalation

Decide beyond tolerance.

Activity 7 Tool Pack

XLSX

48 Risk Assessment Matrix Two pages

Editable evidence file

XLSX

45 Decision Tree Analysis TDIDT

Editable evidence file

XLSX

49 Tornado Chart - Sensitivity Analysis

Editable evidence file

DECISION ENABLED

A prioritised risk register, response and trigger plan, EMV recommendation, tornado sensitivity reading and reconciled S\$18,000 contingency allocation.

Build the Risk Register and Fund Responses

THE SITUATION

The POS interface may be late, refrigerated stock counts are unreliable, a key trainer has a short availability window and a faster payment option could improve adoption. Contingency reserve is S\$18,000.

WHAT YOU'LL DO

Write cause-event-impact risks, rank exposure, assign owners and triggers, choose responses, compare alternatives using EMV and sensitivity, and allocate contingency.

YOU'LL PRODUCE

A prioritised risk register, response and trigger plan, EMV recommendation, tornado sensitivity reading and reconciled S\$18,000 contingency allocation.

DONE WHEN

Priority risks have owners, triggers and funded responses; the EMV and sensitivity logic reconcile; contingency is allocated to named uncertainty and residual risk remains visible.

Tool: Risk Matrix, Decision Tree and Tornado Analysis

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 7 folder.

Debrief — Activity 7: What It Proves

1

Risk management creates options before triggers become expensive issues.

2

Quantitative analysis sharpens material decisions but does not replace judgement.

3

Acceptance means informed monitoring and contingency, not forgetting the risk.

Topic 04 Recap

Decision

Risk is an uncertain event or condition with a positive or negative effect; an issue has already happened.

Method

Qualitative analysis prioritises quickly; quantitative analysis estimates the range of outcomes and decision value.

Evidence

A prioritised risk register, response and trigger plan, EMV recommendation, tornado sensitivity reading and reconciled S\$18,000 contingency allocation.

Capability

Identify, analyse and manage uncertainty with proportionate, funded action (A4, K4).

DAY 1 CLOSE

End of Day 1

Save Activities 1–7. Day 2 continues the same project from the approved risk-aware baseline.

TOPIC 05 · A5, K5 · LO5

Collaborate with Stakeholders

Identify · analyse · communicate · engage · allocate accountability

Stakeholder engagement is part of delivery.

Every unresolved expectation eventually appears as a delay, rejected deliverable, unapproved invoice or uncontrolled change.

Collaborate with Stakeholders — Core Concepts I

01

A stakeholder can affect, be affected by, or perceive themselves to be affected by the project.

02

The stakeholder register captures interest, influence, expectations, impact, communication needs and engagement strategy.

03

Power-interest mapping guides attention; the salience model adds legitimacy and urgency.

04

The engagement assessment matrix compares current and desired states so the gap becomes actionable.

Collaborate with Stakeholders — Core Concepts II

05

RACI clarifies who is Responsible, Accountable, Consulted and Informed; every deliverable needs one clear Accountable owner.

06

Communication planning matches audience, message, channel, frequency, owner and feedback route.

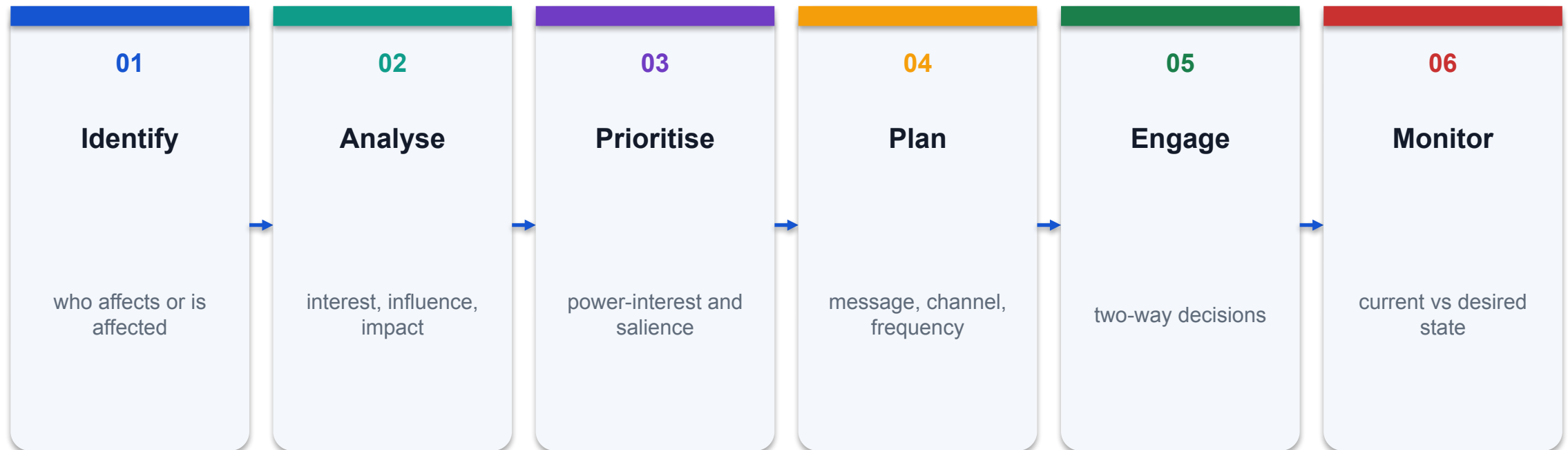
07

Engagement is two-way sense-making and decision participation, not one-way broadcasting.

08

Escalation should state the decision needed, options, impact of delay and recommendation.

Stakeholder Analysis to Engagement



A stakeholder register is useful only when it changes an engagement action.

RACI Diagnostic

Pattern	Meaning	Correction
No A	Nobody owns the outcome	Assign one accountable owner
Multiple A	Competing decision rights	Choose one and define escalation
No R	No one performs the work	Assign delivery responsibility
Too many C	Decision gridlock	Consult only for needed input
Everyone I	Broadcast without purpose	Tailor information needs

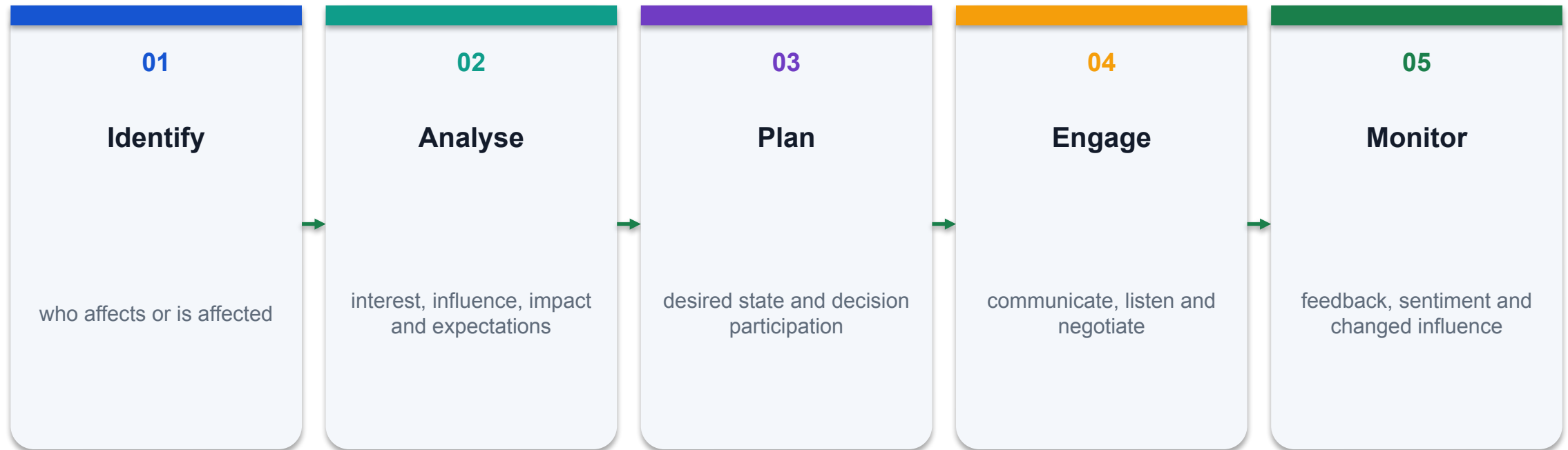
Use deliverables and decisions as rows; vague verbs create vague accountability.

Communication vs Engagement

Dimension	Communication	Engagement
Direction	Often one-way	Two-way
Purpose	Share information	Shape decisions and commitment
Evidence	Message delivered	Feedback changed the work
Risk	Noise and overload	Unresolved expectations

The channel matters less than whether the stakeholder can influence the decision in time.

The Stakeholder Engagement Cycle



Stakeholder identification is continuous because project decisions change who is affected.

Stakeholder Register Fields

Identity

Role, organisation and relationship.

Interest

What outcome matters.

Influence

Formal and informal ability to affect work.

Impact

How the project changes their work.

Expectations

Definition of success and concern.

Strategy

Engagement action and owner.

Stakeholder Mapping Lenses

Lens	Dimensions	Best use
Power-interest	Authority and attention	Management strategy
Impact-influence	Effect and ability to act	Change and adoption
Power-influence	Authority and active influence	Governance
Stakeholder cube	Three dimensions	Complex populations
Salience	Power, legitimacy, urgency	Competing claims

A map is a hypothesis to guide action, not a permanent label.

Seven Salience Classes

Dormant

Power only.

Discretionary

Legitimacy only.

Demanding

Urgency only.

Dominant

Power and legitimacy.

Dangerous

Power and urgency.

Dependent

Legitimacy and urgency.

Definitive

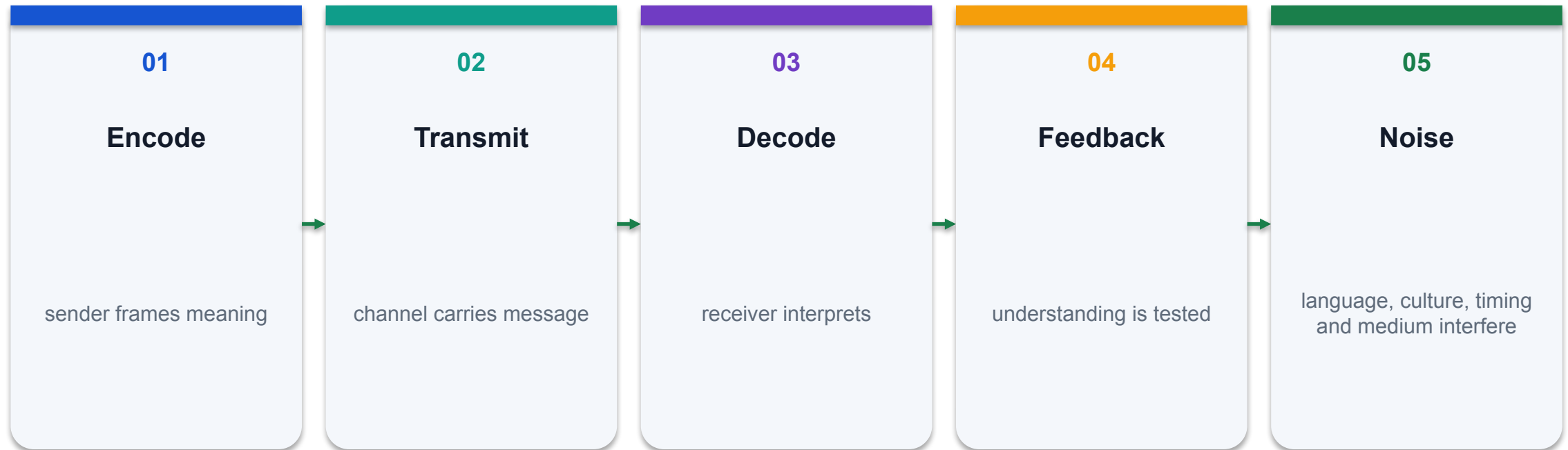
Power, legitimacy and urgency.

Directions of Influence

Direction	Typical stakeholders	Leadership move
Upward	Sponsor and executives	Decision brief and escalation
Downward	Team and specialists	Clarity, coaching and feedback
Sideward	Peers and functional managers	Negotiation and coordination
Outward	Customers, vendors, regulators	Expectation and interface management

Influence without authority depends on evidence, trust and a clear decision request.

Communication Model



Delivery is not communication success; shared understanding is.

Communication Methods

Method	Best use	Risk
Interactive	Decision and ambiguity	Time and facilitation
Push	Targeted distribution	Receipt is not understanding
Pull	Large reference set	Stakeholder may not retrieve it
Visual radiator	Shared current state	Stale information destroys trust

Choose channel based on urgency, complexity, sensitivity and need for feedback.

Stakeholder Engagement Levels

Level	Signal	Management aim
Unaware	Does not know project	Build awareness
Resistant	Opposes change	Understand concern and involve
Neutral	Neither resists nor supports	Connect to relevant value
Supportive	Helps when asked	Sustain participation
Leading	Actively champions	Enable influence

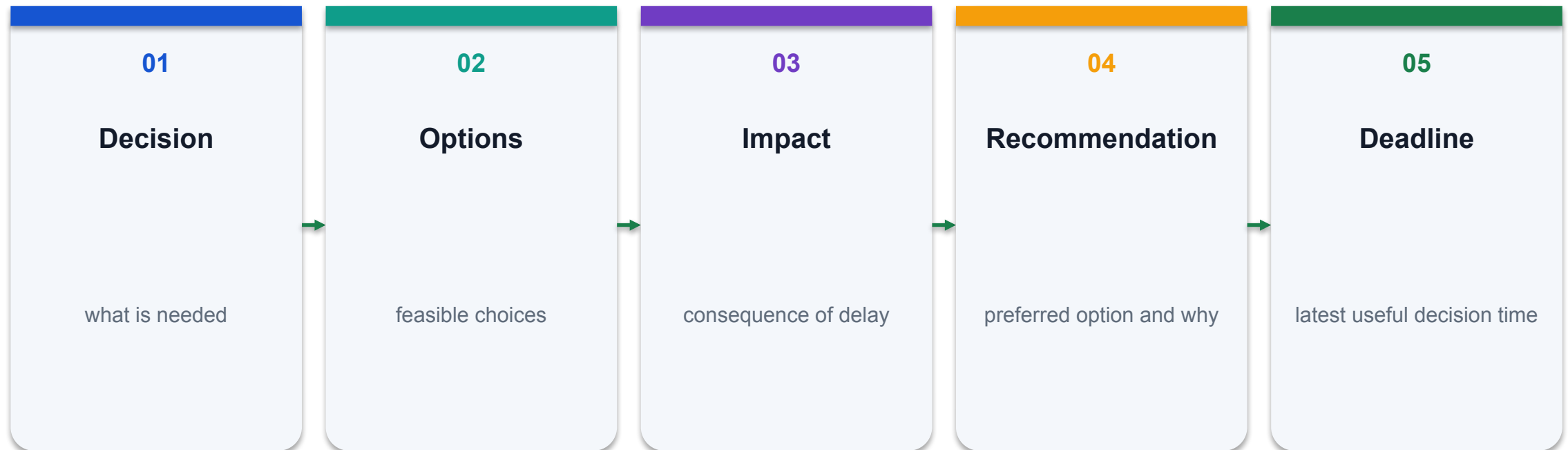
The gap between current and desired state must become an owned action.

HarbourHub RACI Failure Test

Observed row	Failure signal	Repair
Pilot go-live	Sponsor and PM both marked A	Sponsor is sole A; PM remains R
Inventory mapping	Vendor named R with no internal owner	IT lead becomes A; vendor remains R
Outlet job aid	No R assigned	Operations lead owns production
Customer message	Six people marked C	Consult legal and operations only
Benefits review	Everyone marked I	Name finance owner and decision audience

The repaired matrix makes every critical deliverable and decision executable.

Evidence-Based Escalation



Escalation transfers a decision, not a problem without analysis.

Activity 8 Tool Pack

42 Stakeholder Impact Over Influence Matrix

XLSX · editable project tool

43 Salience Chart

PPTX · editable project tool

39 RACI Template

XLSX · editable project tool

41 Stakeholder Engagement Assessment Matrix

XLSX · editable project tool

Align Stakeholders, Accountability and Communications

THE SITUATION

The POS vendor and Operations each believe the other owns interface acceptance. Finance has not approved a payment milestone, store managers receive conflicting instructions and one influential supervisor resists the pilot.

WHAT YOU'LL DO

Map influence and salience, diagnose engagement gaps, clarify RACI, design a two-way communication rhythm and prepare an evidence-based escalation brief.

YOU'LL PRODUCE

A stakeholder map, salience reading, validated RACI, engagement assessment, communication actions and one-minute escalation recommendation.

DONE WHEN

Critical decisions have one Accountable owner, material engagement gaps have two-way actions and the escalation asks the authorised person for a clear, time-bound decision.

Tool: Stakeholder Map, Salience Model, RACI and Engagement Assessment

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 8 folder.

Debrief — Activity 8: What It Proves

1

Many communication failures begin as unclear decision rights.

2

Engagement is participation and feedback, not message broadcasting.

3

Attention follows influence, impact and urgency—not title alone.

Topic 05 Recap

Decision

A stakeholder can affect, be affected by, or perceive themselves to be affected by the project.

Method

The engagement assessment matrix compares current and desired states so the gap becomes actionable.

Evidence

A stakeholder map, salience reading, validated RACI, engagement assessment, communication actions and one-minute escalation recommendation.

Capability

Collaborate with stakeholders and clarify decision rights for execution (A5, K5).

TOPIC 06 · A6 · LO6

Deploy Resources

People · budget · equipment · procurement · team performance

Capacity—not headcount—delivers the plan.

A seven-person team may have only three usable person-days at the constraint. Resource deployment is the discipline of matching skills and time to the critical work.

Deploy Resources — Core Concepts I

01

Resource planning links each work package to required skills, quantities, timing and availability.

02

A responsibility assignment is not a capacity plan; availability must be tested against concurrent work.

03

Resource levelling changes dates to resolve over-allocation; smoothing stays within available float.

04

Fast tracking overlaps work and raises risk; crashing adds cost to critical-path work and may not shorten the project.

Deploy Resources — Core Concepts II

05

Procurement decisions consider make-or-buy, lead time, quality, interfaces, contract type and vendor capability.

06

Tuckman's stages require adaptive leadership: direct, coach, support and delegate as the team matures.

07

Conflict is normal; collaborate for important relationships and outcomes, compromise when time is limited, and escalate only with evidence.

08

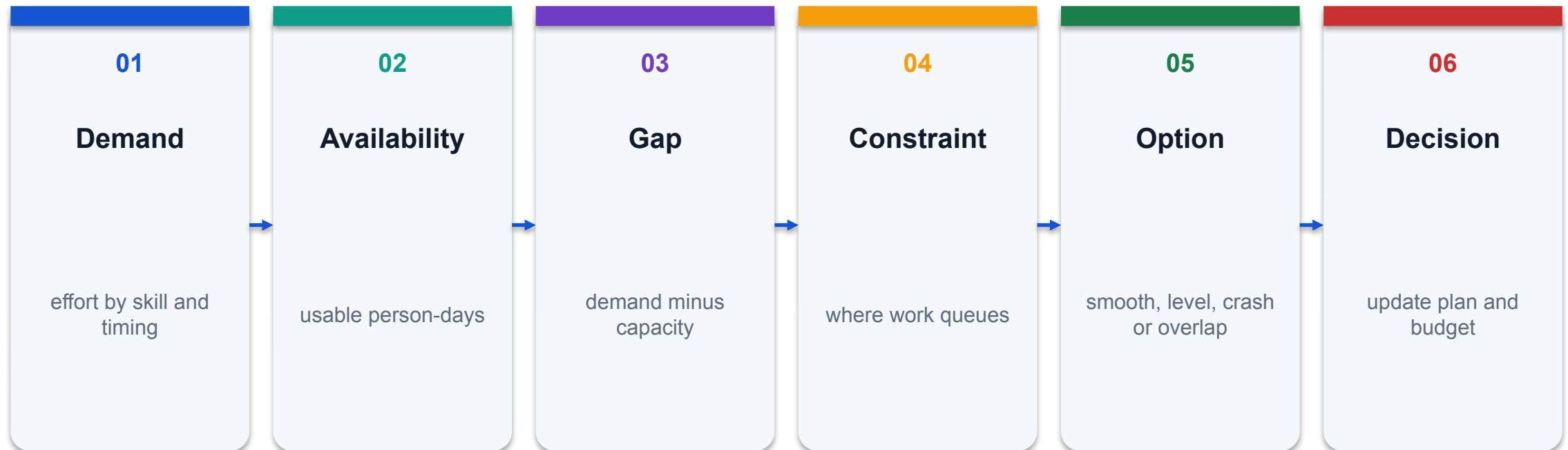
A resource dashboard should show demand, capacity, variance, bottlenecks and decisions—not just headcount.

Four Resource Responses

Option	Schedule effect	Cost / risk effect
Smoothing	Uses float; finish unchanged	Low cost, limited relief
Levelling	May delay finish	Resolves over-allocation
Crashing	Shortens critical work	Adds cost; diminishing returns
Fast tracking	Overlaps sequence	Adds rework and quality risk

Apply compression only to the current critical path and reanalyse after the change.

Capacity Planning



Capacity is time and skill together; headcount alone is not a usable measure.

Team Stage and Leadership

Stage	Team signal	Leader response
Forming	Unclear roles and dependence	Direct and clarify
Storming	Conflict and boundary testing	Coach and facilitate
Norming	Shared standards emerge	Support and reinforce
Performing	Autonomous delivery	Delegate and protect
Adjourning	Transition and loss	Recognise and close

Using one leadership style for every stage creates either dependence or frustration.

Resource Management Plan

Identify

People, equipment, facilities and funding.

Estimate

Quantity, skill and timing.

Acquire

Internal assignment or procurement.

Develop

Capability, trust and working agreements.

Manage

Performance, conflict and feedback.

Control

Usage, availability and corrective action.

Organisation Structure and PM Authority

Structure	PM authority	Resource pattern
Functional	Low	Functional manager controls
Weak matrix	Limited	Coordinator role
Balanced matrix	Shared	Negotiated authority
Strong matrix	Moderate to high	Dedicated PM authority
Project-oriented	High	Dedicated team

Even a small project needs explicit escalation when formal authority is limited.

Role, Authority, Responsibility and Competence

Element	Question	Evidence
Role	What function is performed?	Role description
Authority	What decisions may be made?	Charter and tolerance
Responsibility	What outcome or work is owned?	RACI and work package
Competence	Can the person perform it?	Skill and experience

Assigning work without authority or competence creates hidden dependency.

Two-Week Capacity Worked Example

Skill	Demand	Usable capacity / decision
Integration developer	96 hours	64 hours — defer reporting interface
Outlet trainer	40 hours	48 hours — retain buffer
Data analyst	52 hours	32 hours — buy 20 specialist hours
Test lead	60 hours	56 hours — smooth four hours into next week

The constraint is skill-specific usable time, not the team's nominal headcount.

Acquire, Develop or Procure

Choice	Use when	Trade-off
Internal assignment	Capability exists	Priority conflict
Develop skill	Need is recurring	Learning time
Contract resource	Short-term specialist gap	Cost and knowledge transfer
Outsource deliverable	Clear result and interface	Control and vendor dependency

Choose the smallest intervention that resolves the critical constraint.

Make-or-Buy Decision

Criterion	Make	Buy
Capability	Builds internal skill	Accesses specialist skill
Lead time	Depends on availability	Depends on procurement and onboarding
Control	Direct management	Contract and interface control
Cost	Fixed or opportunity cost	Price plus vendor management
Knowledge	Retained internally	Requires transfer plan

Lowest price is not lowest total project cost when delay and interface risk are included.

Contract Type and Risk

Type	Cost risk	Best fit
Firm fixed price	More seller risk	Stable, clear scope
Cost reimbursable	More buyer risk	Uncertain scope with control
Time and materials	Shared / capped	Short-term flexible effort
Incentive arrangement	Shared performance target	Measurable outcome

The contract allocates financial risk; it does not eliminate delivery accountability.

HarbourHub Team-Stage Diagnostic

Observed behaviour	Stage diagnosis	Next leadership move
Vendor and IT dispute interface ownership	Storming	Facilitate roles and decision rule
Outlet champions share one job aid	Norming	Reinforce standard and peer coaching
Test team resolves defects without escalation	Performing	Delegate within tolerance
Pilot staff worry about release	Adjourning	Recognise contribution and confirm transition

Leadership changes with team evidence; it is not a fixed personal style.

Conflict Approaches

Approach	When useful	Trade-off
Collaborate	Outcome and relationship both matter	Time and openness
Compromise	Time-limited middle ground	Partial satisfaction
Accommodate	Relationship or learning matters more	Your need deferred
Direct / force	Emergency or authority decision	Commitment risk
Avoid	Low value or cooling period	Issue may persist

Conflict is information; choose the response based on stakes and relationship.

HarbourHub Bottleneck Options

Option tested	Schedule effect	Decision
Smooth outlet training	Uses two days of float	Accept — no milestone impact
Level integration work	Moves launch by one week	Reject unless scope is traded
Crash data validation	Recovers three days for S\$3,600	Approve inside reserve
Fast-track testing	May recover four days	Reject — high rework and quality risk

The selected combination protects launch without hiding cost or quality trade-offs.

Team Performance Signals

Delivery

Commitments met with acceptance evidence.

Flow

WIP, blockers and ageing visible.

Quality

Defects and rework trend.

Collaboration

Handoffs, feedback and conflict resolved.

Learning

Capability and knowledge transfer.

Sustainability

Workload remains safe and realistic.

Activity 9 Tool Pack

40 Organisation Chart

PPTX · editable project tool

30 Kanban Board

XLSX · editable project tool

32 Project Budget - Planned vs Actual - Weekly and Monthly

XLSX · editable project tool

09 Resource Capacity and Procurement Decision

XLSX · editable project tool

Deploy Resources and Resolve the Bottleneck

THE SITUATION

Testing needs 22 person-days in two weeks, but only 16 are available. The business analyst is assigned to testing, training and vendor clarification. The sponsor wants evidence before accepting delay or extra cost.

WHAT YOU'LL DO

Map authority and skills, calculate demand-capacity gaps, test smoothing, levelling, crashing and fast tracking, compare make/buy options and recommend a funded deployment decision.

YOU'LL PRODUCE

An organisation map, capacity heatmap, constraint-aware Kanban, procurement comparison, compression options, budget impact and sponsor recommendation.

DONE WHEN

Capacity reconciles to named work and skills, the bottleneck is explicit, procurement/compression options show complete trade-offs and the chosen deployment remains within authorised constraints.

Tool: Organisation Chart, Resource Capacity, Procurement Decision, Kanban and Budget

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 9 folder.

Debrief — Activity 9: What It Proves

1

A RACI assignment does not create capacity.

2

Manage the constraint before adding capacity elsewhere.

3

Crashing buys time with cost; fast tracking buys time with risk.

Topic 06 Recap

Decision

Resource planning links each work package to required skills, quantities, timing and availability.

Method

Fast tracking overlaps work and raises risk; crashing adds cost to critical-path work and may not shorten the project.

Evidence

An organisation map, capacity heatmap, constraint-aware Kanban, procurement comparison, compression options, budget impact and sponsor recommendation.

Capability

Match skills, capacity, procurement and budget to critical work (A6).

TOPIC 07 · A7 · LO7

Track Project Deliverables and Correct Misalignments

Performance · earned value · change · corrective action · handover · closure

Control means deciding early enough to change the outcome.

Variance reporting without forecast, ownership and action is history. Good control protects value and makes closure evidence-ready.

Track Project Deliverables and Correct Misalignments — Core Concepts I

01

Work performance data becomes information only after comparison with the approved baseline and thresholds.

02

A status report should answer: where are we, why, what is forecast, what decision is needed, and who owns the action.

03

Earned value integrates scope, schedule and cost: PV is planned work, EV is completed value, and AC is actual spend.

04

$CPI = EV/AC$ and $SPI = EV/PV$; indices below 1.00 signal inefficiency against the baseline.

Track Project Deliverables and Correct Misalignments — Core Concepts II

05

A change request states reason, impact, options and recommendation; approved changes update baselines and traceability.

06

Corrective action brings performance back to plan; preventive action reduces the probability of future variance.

07

Schedule compression is applied to the current critical path and re-analysed after every change.

08

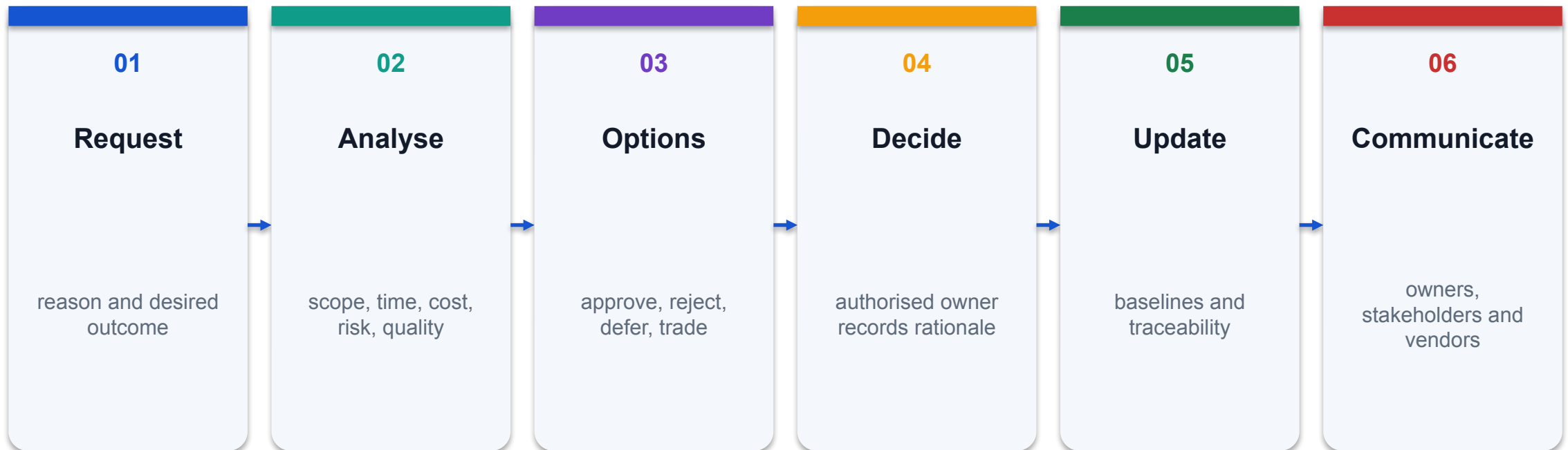
Closure confirms acceptance, transition, contract completion, records, benefits ownership and lessons learned.

Earned Value at Week 12

Measure	Formula / value	Meaning
PV	S\$108,000	Budgeted work planned
EV	S\$90,000	Budgeted value completed
AC	S\$105,000	Actual cost incurred
SPI	$EV/PV = 0.83$	Behind schedule
CPI	$EV/AC = 0.86$	Cost inefficient

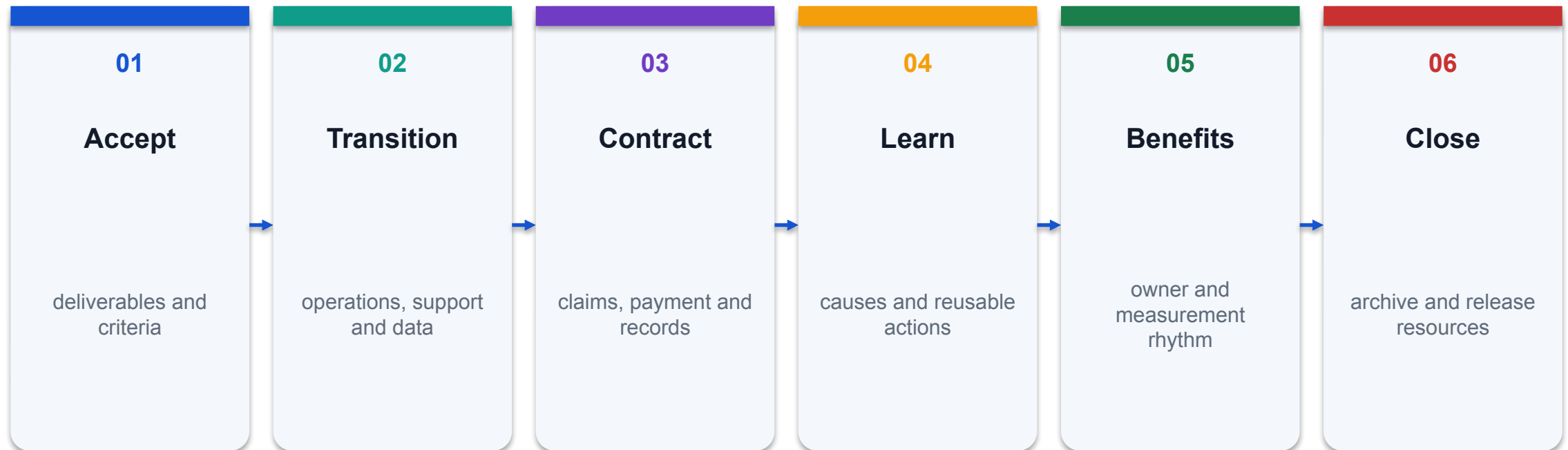
Use the indices with milestones and quality evidence—no single metric tells the whole story.

Integrated Change Control



An approved change is unfinished until affected baselines and acceptance evidence are updated.

Closeout Gates



Do not close on elapsed time; close on complete evidence and transferred accountability.

The Monitoring and Control Cycle



Control is forward-looking: the purpose of variance is to improve the forecast and decision.

Data, Information and Reports

Level	Example	Use
Data	7 defects, S\$105k actual cost	Raw observation
Information	Defects above threshold, CPI 0.86	Meaning through comparison
Report	Forecast, cause, action and decision	Governance and coordination

A dashboard without interpretation is data display, not project control.

A Decision-Ready Status Report

Where

Current milestone and acceptance state.

Variance

Scope, time, cost, quality and resource.

Why

Cause, not just symptom.

Forecast

Expected finish and cost.

Action

Owner, due date and verification.

Decision

Escalation with options and deadline.

Validate Scope versus Control Scope

Process	Question	Evidence
Validate scope	Will the authorised stakeholder accept it?	Accepted deliverable
Control scope	Has authorised scope changed?	Baseline, RTM and change log
Control quality	Does the result conform?	Inspection and test record

Verified quality is not the same as formal customer acceptance.

Earned Value Building Blocks

PV

Budgeted value of work planned by status date.

EV

Budgeted value of work actually completed.

AC

Actual cost of completed work.

BAC

Total authorised budget for measured scope.

Status date

The same cut-off for every measure.

Variance and Performance Indices

Measure	Formula	Interpretation
SV	$EV - PV$	Negative means less value completed than planned
CV	$EV - AC$	Negative means completed value cost more than budgeted
SPI	$EV \div PV$	Below 1.00 means schedule inefficiency
CPI	$EV \div AC$	Below 1.00 means cost inefficiency

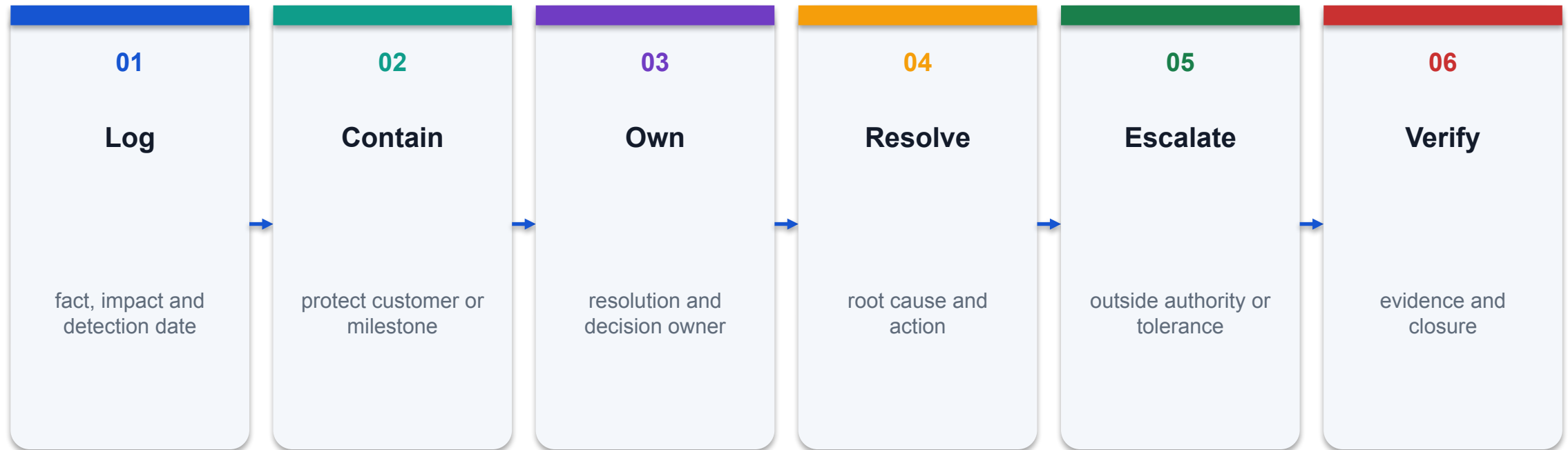
Translate every index into operational consequence and required decision.

Forecasting with EAC

Situation	Formula	Assumption
Current cost efficiency continues	$EAC = BAC \div CPI$	Past cost performance persists
Remaining work re-estimated	$EAC = AC + \text{bottom-up ETC}$	New detailed forecast is credible
Original variance is one-off	$EAC = AC + (BAC - EV)$	Future returns to plan
Cost and schedule both constrain	$EAC = AC + (BAC - EV)/(CPI \times SPI)$	Both inefficiencies persist

Forecast choice is a management assumption and must be stated.

Issue Management



Do not keep an issue disguised as a risk after it has occurred.

Loyalty-Points Change Worked Example

Analysis lens	Finding	Implication
Value	Could lift repeat purchase	Benefit is plausible but unvalidated
Scope and schedule	Adds API and two test cycles	Launch slips nine working days
Cost and resource	S\$18,000 plus scarce developer	Exceeds PM tolerance
Risk and quality	New data/privacy interface	Needs security and regression evidence
Decision	Defer to post-pilot release	Protect baseline; retain request and rationale

Useful ideas are not automatically urgent: the sponsor protects the pilot while preserving the option.

Three Types of Action

Action	Purpose	Example
Corrective	Bring future performance back to plan	Add test capacity to recover milestone
Preventive	Reduce probability of future variance	Add interface review before integration
Defect repair	Modify non-conforming deliverable	Correct catalogue mapping

Each action needs authority, ownership, due date and verification.

Control Across Knowledge Areas

Schedule

Critical path, float and milestone forecast.

Cost

Actuals, earned value and forecast.

Quality

Conformance, variation and corrective action.

Resources

Utilisation, capacity and bottlenecks.

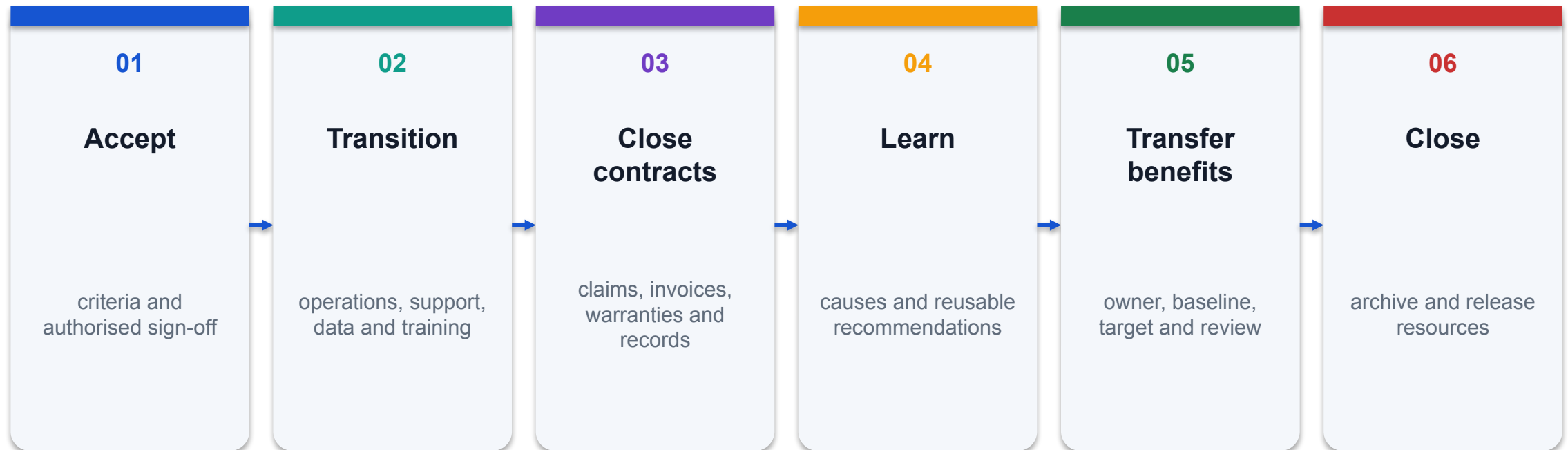
Risk

Triggers, exposure and reserve.

Communication & procurement

Effectiveness, obligations and disputes.

Closure from Acceptance to Benefits



The project can close only after unresolved work is completed or formally transferred.

Activity 11 Tool Pack

31 Burndown Chart

XLSX · editable project tool

32 Project Budget - Planned vs Actual - Weekly and Monthly

XLSX · editable project tool

34 Control Chart

XLSX · editable project tool

11 Integrated Status Change and Issue Log

XLSX · editable project tool

Control Performance, Issues and Change

THE SITUATION

At week 12, PV is S\$108,000, EV is S\$90,000 and AC is S\$105,000. The sponsor requests loyalty points for S\$24,000 and three weeks while a vendor interface issue threatens acceptance.

WHAT YOU'LL DO

Calculate earned value and forecast, integrate quality and milestone evidence, triage the issue, analyse the change, obtain an authorised decision and define corrective action.

YOU'LL PRODUCE

An integrated status report, EVM calculations and forecast, issue decision, change-impact analysis, decision log, corrective action and updated control records.

DONE WHEN

EVM and forecast reconcile, issue and change records are distinct, the decision is authorised, all affected records are updated and corrective action is measurable.

Tool: Integrated Status, Change and Issue Log, Burndown, Budget and Control Chart

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 11 folder.

Debrief — Activity 11: What It Proves

1

Status reporting is a decision system, not a colour-coded history lesson.

2

Change control enables deliberate trade-offs; it does not prevent change.

3

An approved change is incomplete until every affected baseline is updated.

Activity 12 Tool Pack

XLSX

44 Agile Retrospective

Editable evidence file

XLSX

12 Acceptance Handover Closure and Benefits

Editable evidence file

DECISION ENABLED

Signed acceptance matrix, handover and training evidence, unresolved-item disposition, contract closeout, lessons register, benefits tracker and formal closure recommendation.

Accept, Handover and Close the Project

THE SITUATION

All three outlets are live, but final payment is requested before complete acceptance evidence, support ownership, training records and unresolved-item decisions are consolidated. Benefits will continue after project closure.

WHAT YOU'LL DO

Validate acceptance, transfer operations and knowledge, resolve contracts and open items, capture lessons and assign post-project benefits measures before recommending closure.

YOU'LL PRODUCE

Signed acceptance matrix, handover and training evidence, unresolved-item disposition, contract closeout, lessons register, benefits tracker and formal closure recommendation.

DONE WHEN

Acceptance is traceable, handover ownership is explicit, open items have formal disposition, contracts and records reconcile, and benefits continue under named operational owners.

Tool: Acceptance, Handover, Closure and Benefits Tracker plus Retrospective

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 12 folder.

Debrief — Activity 12: What It Proves

1

Elapsed time does not close a project; complete evidence and transferred accountability do.

2

Lessons become assets only when future owners can find and use them.

3

Project work ends before benefits measurement ends.

Topic 07 Recap

Decision

Work performance data becomes information only after comparison with the approved baseline and thresholds.

Method

$CPI = EV/AC$ and $SPI = EV/PV$; indices below 1.00 signal inefficiency against the baseline.

Evidence

Signed acceptance matrix, handover and training evidence, unresolved-item disposition, contract closeout, lessons register, benefits tracker and formal closure recommendation.

Capability

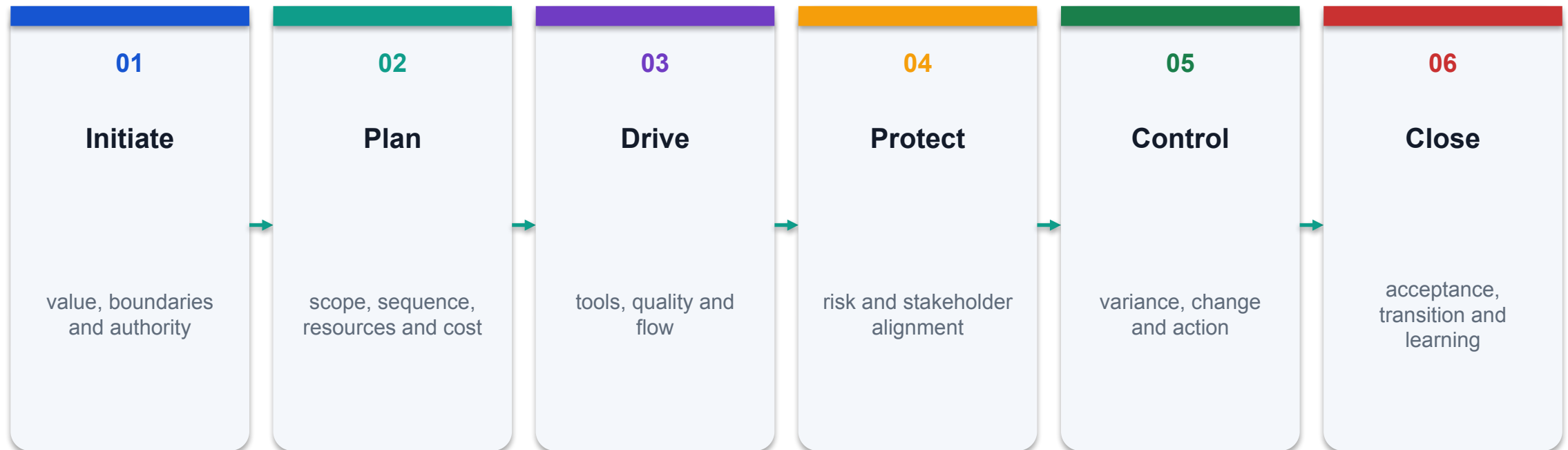
Close only when deliverables, transition, contracts, records and benefits ownership are evidenced (A7).

WRAP-UP

Course Wrap-Up

From authorisation to controlled handover—one connected evidence chain.

The Small-Project Delivery System



Every output becomes an input to the next decision.

Lifecycle Takeaways

01

Start with value and measurable success, not a list of tasks.

02

Integrate scope, schedule and cost around the same work packages.

03

Choose the tool that changes the next decision and retire unused reporting.

04

Fund risk responses before triggers become issues.

05

Make accountability and stakeholder feedback visible.

06

Plan capacity at the constraint, not by nominal headcount.

07

Control through forecast and action; close through evidence and transfer.

Assessment Readiness Check

Knowledge

Can you explain the small-project elements, plan requirements, tools, risks and stakeholder identification?

Application

Can you produce and interpret the charter, baseline, quality analysis, risk register, RACI, resource plan and control evidence?

Judgement

Can you choose a method, explain the trade-off and recommend a proportionate action?

Evidence

Can you trace each recommendation to the case facts and an approved project artifact?

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Thank You!

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